

A Theory of Contact and In-Group Bias*

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Abstract

Why do some contact interventions durably reduce in-group bias while others fail? I develop a dynamic model of bias evolution in which temporary increases in out-group contact generate permanent change only if they push the population past a tipping point. The model delivers four results. First, endogenous effort attenuates the societal feedback sustaining multiplicity, so the parameter region admitting tipping shrinks monotonically as the value of successful interaction rises. Second, value-enhancing interventions can eliminate the high-bias steady state entirely, producing permanent change over a broader range of initial conditions than contact interventions. Third, sequencing—raising the value of interaction first, then increasing contact—can shorten the required duration of the value-enhancing phase. Fourth, a welfare analysis shows that value-enhancing interventions raise welfare on impact while contact interventions impose a front-loaded welfare penalty, so the optimal instrument choice can differ from the one minimizing fiscal cost. The model also predicts a majority-minority asymmetry in steady-state bias driven by differential in-group exposure.

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How can someone hate me, if they don't even know me?

- Daryl Davis

1 Introduction

Contact interventions designed to reduce in-group bias have produced a striking pattern in the empirical literature: structurally similar programs generate sharply different long-run outcomes across settings. Lowe (2021) finds that cooperative cross-caste contact through a cricket program in India generates persistent and broad reductions in discrimination that extend beyond the contact setting. By contrast, Mousa (2020) studies a closely analogous intervention in post-ISIS Iraq, pairing Christians and Muslims through a soccer program, and finds that its effects remain confined to the contact setting. More broadly, other studies find weak or null long-run effects of structured contact interventions (Scacco and Warren, 2018; Clochard et al., 2026), while meta-analyses and reviews conclude that treatment effects are highly heterogeneous across settings (Pettigrew and Tropp, 2006; Paluck et al., 2021; Lowe, 2025). A standard response is to attribute this variation to differences in implementation, population characteristics, or historical context. However, that approach neither yields a general explanation for why contact succeeds in some environments and fails in others nor provides much guidance for policy design.

This paper offers a different explanation. I develop a dynamic model of in-group bias evolution in which the long-run effect of a contact intervention depends on where the population stands relative to an endogenous tipping point. The mechanism implies that even interventions with broadly similar design can produce sharply different persistence across settings. When bias is self-reinforcing at the societal level, the dynamics admit two stable steady states, low bias and high bias, separated by an unstable threshold. A temporary increase in out-group contact generates a permanent reduction in bias only if it moves the population across that threshold; otherwise, bias declines during the intervention and reverts once contact returns to baseline. The same program can succeed in one setting and fail in another, even under similar design, because the model identifies three distinct structural channels through which settings can differ: prevailing bias may place one population within reach of the tipping point but leave another beyond it; the strength of societal feedback may generate tipping dynamics in one environment but not the other; and even when both settings exhibit tipping, differences in that feedback structure can shift the location of the tipping window so that identical initial bias falls inside it in one context and outside it in the other.

There is a population divided into two groups, a majority and a minority, whose members are repeatedly matched for social interaction.¹ The degree of assortative matching governs the frequency of out-group contact and is itself an object of policy. After matching, the probability of a positive interaction depends on three channels: the effort chosen by the matched individuals, their biases, and the aggregate level of bias in the population. Out-group interactions reduce bias when positive and reinforce it when negative, while in-group interactions reinforce bias through repeated in-group contact.² The state variables are the group-specific shares of biased individuals, whose co-evolution determines the long-run dynamics.

The societal feedback channel captures the idea that widespread bias shapes the quality of social interactions beyond the direct effect of the interacting individuals' own biases. When bias is pervasive, social norms and expectations can make positive out-group interaction difficult to sustain even for privately unbiased individuals, who may face external pressure to conform to the norms of their environment. When bias is rare, even privately biased individuals may find their views tempered by surrounding norms of cooperation. This feedback between aggregate bias and individual interaction outcomes is the source of the paper's tipping dynamics and has intuitive support from theories of social norm enforcement (Bowles and Gintis, 2002) and preference falsification (Kuran, 1995).

The paper's central contribution is a theory of when temporary contact generates permanent change. A temporary increase in the out-group contact rate generates a permanent reduction in bias only if the intervention is strong enough to push the population past the tipping point. If the intervention falls short, bias declines during the contact period but reverts once contact returns to baseline. This result is consistent with evidence that short-lived or weakly institutionalized contact interventions often produce weak or null effects (Paluck et al., 2019; Scacco and Warren, 2018; Enos, 2014), while sustained or structured contact can generate lasting change (Bazzi et al., 2019; Lowe, 2021; Ghosh et al., 2026). The mechanism differs from both canonical discrimination models, in which contact does not shift long-run outcomes, and belief-correction models, in which positive contact reduces bias without threshold effects (Becker, 1957; Arrow, 1973; Bordalo et al., 2016; Bohren et al., 2025).³

¹Groups may correspond to latent forms of identity such as ethnicity, political orientation, race, religion, or social class. See, e.g., Akerlof and Kranton (2000, 2010), Chen and Li (2009), Bénabou and Tirole (2011), Carvalho (2016), Akerlof (2017), and Carvalho and Sacks (2021, 2024).

²Examples of in-group interactions reinforcing in-group bias include Tajfel et al. (1971), Tajfel and Turner (1979), Goette et al. (2006), Bernhard et al. (2006), and Chen and Li (2009).

³Related models include Coate and Loury (1993b), who study statistical discrimination, and Bohren et al. (2019), Peşki and Szentes (2013), and Heidhues et al. (2026), who study belief distortions and social norms. In these settings, contact operates through informational or reputational channels rather than through tipping dynamics, and does not deliver the sequencing result below.

Where contact interventions fail, increasing the value of successful interaction can succeed. In the model, this is captured by an increase in V , the payoff from a successful interaction. Thus, V should be interpreted not as contact per se, but as the material or institutional stake attached to successful interaction. Policies that raise V therefore correspond to interventions that make successful interaction more valuable, including economic integration into high-value productive settings, team-based pay, or school environments organized around meaningful shared goals. By raising the return to effort, such policies induce investment in positive interactions even in high-bias environments and thereby weaken the feedback loop that sustains the high-bias steady state. Because long-run bias is shaped by the success or failure of out-group interactions, a higher V reduces bias by making those interactions more likely to succeed. When the returns to successful interaction are sufficiently high, the high-bias steady state disappears during the intervention, leaving a unique low-bias steady state to which the population converges from any initial condition. Once bias falls below the original tipping point, the policy can be withdrawn without undoing the change. Unlike contact interventions, value-based interventions can therefore generate permanent change even in settings where temporary increases in contact would fail.

The asymmetry between the two instruments implies a sequencing result. The instruments are complements rather than substitutes. A sufficiently strong value-based intervention alone can generate permanent change from any initial condition, but may require a prolonged or intensive intervention. A natural sequencing logic therefore combines the two instruments. A temporary value-based intervention first reduces bias enough to move the population into the tipping region, after which a temporary increase in out-group contact can push it across the threshold that contact interventions alone could not have reached. The formal sequencing result further shows that such a policy sequence can shorten the duration of the value-based intervention relative to a value-only policy, though not necessarily the total duration of intervention. Once the economy crosses the baseline tipping point, both interventions can be withdrawn and the low-bias outcome becomes self-sustaining. Moreover, accounting for the welfare effects of each instrument along the transition path shows that the fiscal case for sequencing can overstate its attractiveness. The V -phase periods eliminated by sequencing may themselves have been generating net welfare gains.

This interpretation is consistent with evidence from high-value integration settings. Rao (2019) documents persistent reductions in discriminatory behavior among wealthy students following integration with low-income peers in elite Delhi schools, with the largest effects appearing in economic

exchange tasks where interaction stakes are highest. Boisjoly et al. (2006) find lasting attitude change among white students randomly assigned African American roommates, and Beaman et al. (2009) find persistent reductions in gender bias following repeated exposure to female leaders under a quota system. In each case, the intervention contains features that can be interpreted through the model as raising the value of successful interaction in settings organized around out-group cooperation, rather than merely increasing contact rates.⁴

The effort margin creates a central tension in the model. Effort reduces steady-state bias while weakening the feedback required for permanent regime change. Because individuals respond to rising aggregate in-group bias by investing more effort in out-group interactions, the feedback loop is dampened relative to a model without effort. This attenuation has two consequences. First, effort rules out the fully biased steady state. Even in high-bias environments, individuals invest enough in out-group interactions to keep universal bias from being sustained. Second, effort shrinks the parameter region supporting multiple steady states, so the conditions required for tipping dynamics are harder to satisfy than a model without effort would suggest. Raising the value of successful interaction therefore lowers steady-state bias by inducing investment in effort. However, the same response flattens the feedback and narrows the conditions under which tipping dynamics arise. This tension is absent from standard models of evolutionary dynamics and equilibrium selection (Schelling, 1971; Kandori et al., 1993; Young, 1993). The same force that reduces steady-state bias also attenuates the feedback required for temporary interventions to generate permanent change. The model therefore yields a sharp empirical prediction. Environments with higher stakes for successful interaction should exhibit lower steady-state bias, but a smaller region in which temporary increases in contact generate lasting change. Card et al. (2008) show that tipping points play an important role in the dynamics of residential segregation; the present paper identifies an analogous threshold structure in the evolution of in-group bias, but shows that effort makes that structure harder to sustain. This attenuation result is central to the paper. Adaptation to bias via effort does not merely reduce bias in levels, it also changes which policies can generate permanent change.

The model also yields a sharp asymmetry in steady state in-group bias. Under empirically plausible conditions of non-fully-assortative matching and positive in-group reinforcement, majority members are strictly more biased than minority members in any interior steady state, and minority members invest more effort in out-group interactions than majority members. The mechanism is differential

⁴This is consistent with Allport's (1954) argument that contact is most effective when it occurs under cooperative goals and meaningful shared stakes.

in-group exposure. Because the majority’s in-group interaction rate exceeds the minority’s, majority members accumulate more in-group reinforcement, which shifts their steady-state bias upward. A uniform increase in out-group contact reduces this gap, generating convergence in the distribution of bias across groups alongside a reduction in average bias, consistent with evidence that contact effects are often stronger among majority than minority group members (Tropp and Pettigrew, 2005; Ghosh et al., 2026), and with evidence that higher contact intensity can backfire among majority but not minority participants (Ghosh et al., 2026). This asymmetry is distinct from that in identity-based (Akerlof and Kranton, 2000; Shayo, 2009) or cultural-transmission and homophily models (Bisin and Verdier, 2001; Currarini et al., 2009; Golub and Jackson, 2012), because here it emerges from the interaction structure itself rather than from differences in preferences or socialization across groups.

Relative to existing models of social interactions with aggregate feedback, the paper’s contribution is not simply to show that sufficiently nonlinear social feedback can generate tipping. Rather, it identifies how tipping dynamics are reshaped once out-group interaction is mediated by endogenous effort and match-specific bias. The model delivers four results that do not arise from the societal-feedback channel alone. First, endogenous effort attenuates the aggregate feedback that sustains multiplicity, so the same force that lowers steady-state bias also makes permanent regime change harder to achieve. Second, temporary contact and value-enhancing interventions operate through distinct mechanisms. The former works, when it works, by shifting the tipping point and changing basins of attraction, whereas the latter can eliminate the high-bias steady state altogether by raising the return to effort. Third, this asymmetry implies a formal sequencing result. A temporary value-enhancing intervention can move the economy into the region where contact becomes self-sustaining, and can shorten the required duration of the value-enhancing phase relative to relying on that intervention alone. In the effort-only benchmark, the dynamics admit a unique steady state. In the societal-feedback benchmark, sufficiently sigmoidal feedback generates multiplicity and tipping. In the full model, endogenous effort dampens the aggregate feedback that sustains multiplicity, shrinking the region in which temporary increases in contact can generate permanent change. As V rises, this attenuation becomes stronger, and the multiplicity region can disappear altogether. Figure 5 illustrates these results. Fourth, a welfare analysis shows that value-based interventions raise welfare on impact while contact-based interventions impose a front-loaded welfare penalty, so that accounting for transition-path welfare tilts the optimal policy choice toward value-enhancing interventions relative to what fiscal cost alone would suggest.

The remainder of the paper is structured as follows. Section 2 develops the model and establishes preliminary results. Sections 3 and 4 form the paper's analytical core: the former isolates each mechanism individually, while the latter combines them to characterize the full dynamics. Section 5 derives the welfare effects of each instrument and determines the efficient policy choice. Section 6 discusses policy implications, and Section 7 concludes.

2 The Model

Time is discrete and indexed by $t = 0, 1, 2, \dots$. There is a fixed population of individuals $I = [0, 1]$ with Lebesgue measure λ . Each $i \in I$ is born with a type $\tau \in \{1, 2\}$. $I_\tau \subset I$ denotes the subset of τ types. The share of type 1 individuals is $q = \lambda(I_1) > \frac{1}{2}$, so type 1 individuals constitute the majority. Generically, $q_\tau = \lambda(I_\tau)$. Each $i \in I$ is also endowed with an in-group bias parameter $b_i^t \in \{0, 1\}$. If $b_i^t = 0$, then i is *unbiased* and if $b_i^t = 1$, then i is *biased*. These biases evolve over time in a manner described below. Let $\pi_\tau^t = \frac{1}{\lambda(I_\tau)} \int_{I_\tau} b_i^t d\lambda \in [0, 1]$ denote the share of biased type τ individuals in the population at time t and $\pi^t = (\pi_1^t, \pi_2^t)$. The initial state π^0 is exogenously given. Let $\bar{\pi}^t = q\pi_1^t + (1 - q)\pi_2^t$ denote the aggregate prevalence of in-group bias in the population. When there is no confusion, I suppress the time notation.

Timing. Social interactions in each period take place as follows.

- (1) Each individual i is matched with an individual j , at which point (τ_i, b_i) is observed by j and (τ_j, b_j) is observed by i . Let

$$\eta_\tau = \lambda(I_\tau) - \mu(1 - \lambda(I_\tau))$$

denote the probability that a type τ individual is matched with another type τ individual.⁵ The *match parameter* $\mu \in [-1, \frac{1-q}{q}]$ guarantees that $\eta_\tau \in [0, 1]$ for $\tau = 1, 2$. Random matching occurs when $\mu = 0$, assortative matching occurs when $\mu < 0$, and disassortative matching occurs when $\mu > 0$.

- (2) Upon individuals i and j being matched, each chooses an effort level $e \in [0, 1]$ at cost $\frac{1}{2}e^2$ to maximize current utility, described below.⁶ Their efforts e_i and e_j are aggregated according

⁵This specification satisfies market clearing: $q(1 - \eta_1) = (1 - q)(1 - \eta_2)$; i.e., the share of type 1 individuals matched with type 2 individuals must equal the share of type 2 individuals matched with type 1 individuals.

⁶The qualitative results are invariant to whether effort is chosen before or after matching. Under pre-match timing, individuals form expectations over the type-bias distribution of their match, yielding a Bayesian Nash equilibrium characterized by a 4×4 linear system with the same properties as Proposition 1.

to the function

$$s(e_i, e_j) = e_i + e_j - e_i e_j, \quad (1)$$

which determines the probability that effort alone yields a positive interaction, described next.⁷

- (3) The social interaction between i and j occurs. Three factors determine whether this interaction is positive or negative.

- (i) *Effort.* With probability $s(e_i, e_j)$, the interaction is positive.

If effort is not sufficient to yield a positive interaction, then the biases of the matched individuals and the broader social environment influence the outcome.

- (ii) *Individual feedback.*

If effort fails to yield a positive interaction, occurring with probability $1 - s(e_i, e_j)$, then the matched individuals' biases determine the probability that the interaction is resolved positively at the next stage. Given biases b_i and b_j , let

$$\beta(b_i, b_j) = \begin{cases} \beta_0 & \text{if } b_i + b_j = 0 \\ \beta_1 & \text{if } b_i + b_j = 1 \\ \beta_2 = 0 & \text{if } b_i + b_j = 2, \end{cases} \quad (2)$$

where $0 = \beta_2 < \beta_1 < \beta_0 \leq 1$.⁸

For in-group matches, I suppress the bias notation and treat the matched pair as if $(b_i, b_j) = (0, 0)$, since the individual-bias channel is only relevant for out-group interactions.

⁷The specific functional form $s(e_i, e_j) = e_i + e_j - e_i e_j$ is adopted for tractability rather than necessity. The results extend to any function $s : [0, 1]^2 \rightarrow [0, 1]$ that is increasing and concave in each player's own effort with effort choices as strategic substitutes. Under these conditions, equilibrium effort is increasing in $(1 - \beta(b_i, b_j))\theta(\bar{\pi})V$, which is the only way s enters the equilibrium comparative statics and the fixed-point structure of the dynamics. The assumption that effort choices are strategic substitutes reflects the standard free-riding incentive in team and pair-based production (Holmström, 1982). This interpretation is consistent with field evidence showing that team productivity in socially divided workplaces depends sensitively on how incentives are structured (Hjort, 2014).

⁸ $\beta_0 > 0$ implies that, even in the absence of effort, unbiased individuals extend some baseline trust and willingness to engage in interaction (Berg et al., 1995; Glaeser et al., 2000). The assumption $\beta_1 < \beta_0$ reflects the idea that bias reduces the probability of successful coordination by distorting expectations and incentives in out-group interaction (Arrow, 1973; Akerlof and Kranton, 2000). Finally, $\beta_2 = 0$ captures a limiting case in which mutual bias produces complete distrust and interaction breakdown, consistent with theories of out-group prejudice and discrimination (Tajfel and Turner, 1979; Becker, 1957; Coate and Loury, 1993a). Related models show how beliefs about group characteristics can generate persistent discrimination and inefficient interaction outcomes (Arrow, 1973; Coate and Loury, 1993b; Bohren et al., 2019, 2025). $\beta_2 = 0$ is adopted for convenience, not necessity. For any $\kappa \in (0, \beta_1)$ $\rho_{1,1} > 0$ and the fully biased steady state shifts by continuity to an interior point at distance $O(\kappa)$ from $(1, 1)$. All qualitative results are unchanged.

(iii) *Societal feedback.* If neither effort nor overcoming individual biases are sufficient to yield a positive interaction, occurring with probability $(1 - s(e_i, e_j))(1 - \beta(b_i, b_j))$, the outcome of the interaction is determined by the broader social environment. Let $\bar{\pi}$ denote the average level of bias in the population. Conditional on reaching this stage, the probability that the interaction fails is given by $\theta(\bar{\pi})$, where $\theta : [0, 1] \rightarrow [0, 1]$ is increasing and twice continuously differentiable with $\theta(0) = 0$ and $\theta(1) = 1$. I interpret $\theta(\bar{\pi})$ as a reduced form model of societal feedback that captures the idea that when bias is widespread, social norms and expectations make successful interaction less likely even after conditioning on the characteristics of the interacting pair. Section 2.1 offers two microfoundations that yield this reduced form specification.

Note that the paper's results are invariant to the ordering of (i)–(iii).⁹

- (4) Biases are updated as follows. If $\tau_i = \tau_j$ at time t , then with probability $\varepsilon > 0$, $b_i^{t+1} = 1$ and with complementary probability, $b_i^{t+1} = b_i^t$. If $\tau_i \neq \tau_j$ at time t , then a positive interaction yields $b_i^{t+1} = 0$ and a negative interaction yields $b_i^{t+1} = 1$.¹⁰

Hence, conditional on i and j being matched, the probability of a positive interaction is

$$\rho_{b_i, b_j}(e_i, e_j) = s(e_i, e_j) + (1 - s(e_i, e_j))[\beta(b_i, b_j) + (1 - \beta(b_i, b_j))(1 - \theta(\bar{\pi}))]. \quad (3)$$

This process is illustrated in Figure 1.

Payoffs. Conditional on being matched with an individual j with type-bias pair (τ_j, b_j) , individual i 's utility is

$$u(e_i, e_j; (\tau_i, b_i), (\tau_j, b_j)) = \rho_{b_i, b_j}(e_i, e_j)V - \frac{1}{2}e_i^2, \quad (4)$$

where $V > 0$ is the value received from a successful interaction. Throughout, a positive interaction is the payoff-relevant successful outcome of the interaction. A high V implies high interaction stakes and a low V implies low interaction stakes.

Equilibrium Definitions. At each time t , the utility-maximizing type-and-bias contingent effort levels constitute a Nash equilibrium (NE) of the social interaction stage game if, for all i , $e(\tau_i, b_i) =$

⁹As $\rho_{b_i, b_j}(e_i, e_j) = s(e_i, e_j) + (1 - s(e_i, e_j))[\beta(b_i, b_j) + (1 - \beta(b_i, b_j))(1 - \theta(\bar{\pi}))]$, which can be written as $1 - (1 - s)(1 - \beta)\theta$, all six orderings of the three channels are identical (as this probability is symmetric in effort, individual feedback, and societal feedback).

¹⁰I discuss the construction of ε in greater detail in Section 2.2.

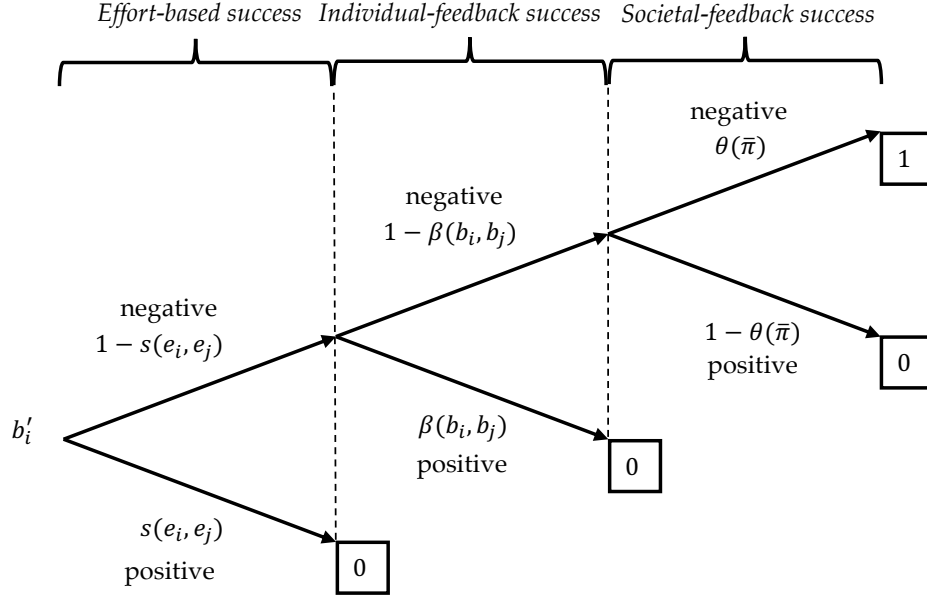


Figure 1: Diagrammatic representation of the probability of a positive interaction and future bias b'_i given current bias b_i and the matched individual's bias b_j . *Positive* corresponds to a positive interaction and *negative* to a negative interaction.

$\arg \max_{e_i \in [0,1]} u(e_i, e(\tau_j, b_j); \tau_i, b_i, \tau_j, b_j)$. This equilibrium describes within-period strategic interactions. Across periods, the distribution of biases evolves according to the process described above.

Let $\pi^\infty = (\pi_1^\infty, \pi_2^\infty)$ denote the steady-state shares of in-group bias among each type and $\bar{\pi}^\infty$ the steady-state aggregate bias. I define two steady states of interest. The biased steady state is the steady state in which the entire population is biased ($\bar{\pi}^\infty = 1$), and the unbiased steady state is the steady state in which the entire population is unbiased ($\bar{\pi}^\infty = 0$).

2.1 Microfoundations of the Societal-Feedback Channel

While the model treats $\theta(\bar{\pi})$ as a reduced-form object, it can be backed out from structural models of societal interactions. The microfoundation discussed here is community norm enforcement. Conditional on reaching the societal-feedback stage, the probability of a successful interaction is mediated by the biases of a finite reference group drawn from the population.

Consider a model of community norm enforcement. Conditional on reaching the societal-feedback stage, the interaction is mediated by a reference group of k members drawn independently from the population. If a majority of the k members drawn are biased, then the social interaction is deemed

a failure and if a majority are unbiased, then the social interaction is successful. Thus,

$$\theta(\bar{\pi}) = \Pr \left(\text{Binomial}(k, \bar{\pi}) > \frac{k}{2} \right) = \sum_{j=\lfloor k/2 \rfloor + 1}^k \frac{k!}{j!(k-j)!} \bar{\pi}^j (1 - \bar{\pi})^{k-j}.$$

For all k , $\theta(0) = 0$ and $\theta(1) = 1$. At $k = 1$, $\theta(\bar{\pi}) = \bar{\pi}$ and at $k = 2$, $\theta(\bar{\pi}) = \bar{\pi}^2$.¹¹

For all $k \geq 3$, θ is sigmoidal, and $\sup_{\bar{\pi}} \frac{d\theta(\bar{\pi})}{d\bar{\pi}}$ is strictly increasing in k . Thus, larger reference groups generate sharper threshold behavior in societal feedback. Moreover, as $k \rightarrow \infty$, $\theta(\bar{\pi})$ converges pointwise to a step function at $\bar{\pi} = \frac{1}{2}$, so the size of the relevant social reference group governs the strength of societal feedback. The formal link between this curvature and the existence of multiple steady states is developed below.¹²

A preference-falsification environment in the spirit of Kuran (1995) can also generate the reduced form specification above. These microfoundations capture the broader idea, emphasized in the literature on trust and social capital, that the social environment shapes the expressed quality of bilateral interactions (Bowles and Gintis, 2002) and are illustrative rather than essential. They offer structural environments in which the reduced-form societal-feedback channel can arise, but none of the analytical results depends on any specific functional form or underlying mechanism of social interaction. The proofs are stated for a general reduced-form $\theta(\bar{\pi})$.

2.2 Preliminary Results

Before characterizing the full dynamics, I establish three preliminary results. I first solve for the effort levels that constitute a Nash equilibrium of the social interaction stage game. I then characterize the implied average effort gap between minority and majority individuals. Lastly, I highlight the role of randomness and in-group reinforcement in the evolution of bias.

Recall that, conditional on being matched with a type τ_j individual with bias b_j , individual i chooses the effort level that maximizes (4).

Lemma 1. *Suppose that pairs (τ_i, b_i) and (τ_j, b_j) are matched. The effort levels constituting a Nash*

¹¹For even k , ties arise with positive probability. Any tie-breaking rule preserves the qualitative conclusions emphasized here. $\theta(\bar{\pi})$ remains increasing, $k = 1$ yields linear feedback, $k = 2$ yields convex feedback, and larger k produces increasingly sharp threshold behavior. The exact location of the inflection point may depend on how ties are resolved, but none of the analytical results relies on that detail.

¹²The conditions for multiple steady-states are outlined in Proposition 2. When the remaining conditions of Proposition 2 are satisfied, there always exists a k large enough to generate multiple steady states and tipping dynamics.

equilibrium of the social interaction stage game are given by

$$e_i^*(b_i, b_j) = \frac{(1 - \beta(b_i, b_j))\theta(\bar{\pi})V}{1 + (1 - \beta(b_i, b_j))\theta(\bar{\pi})V}. \quad (5)$$

Thus, effort levels are increasing in the value of a successful interaction, the match-specific probability of failure, and the societal-feedback function $\theta(\bar{\pi})$.

The proofs of this and all other results are in the Appendix. Lemma 1 implies that equilibrium effort is increasing in the match-specific probability of a negative interaction, $1 - \beta(b_i, b_j)$, the societal-feedback term $\theta(\bar{\pi})$, and the value of a successful interaction V . Interpreting effort as costly preventive investment, individuals invest more when baseline success is less likely. This logic is consistent with evidence that cooperation rises when punishment or exclusion makes opportunism more consequential (Fehr and Gächter, 2000; Cinyabuguma et al., 2005; Gürer et al., 2006). Related field evidence shows that incentive design, including team-based pay, can mitigate productivity losses associated with inter-group divisions (Hjort, 2014).

Lemma 1 also allows a comparison of average effort between minority (type 2) and majority (type 1) individuals. Given (5), the average effort put forth by a type τ is

$$(1 - \eta_\tau)[\pi_1\pi_2e^*(1, 1) + ((1 - \pi_1)\pi_2 + (1 - \pi_2)\pi_1)e^*(1, 0) + (1 - \pi_1)(1 - \pi_2)e^*(0, 0)] + \eta_\tau e^*(0, 0),$$

where $e_i^*(b_i, b_j) = e^*(b_i, b_j)$ as i enters only through b_i and $e^*(1, 0) = e^*(0, 1)$.

Corollary 1. *Average effort is weakly higher among minority individuals than majority individuals, strictly so when $\mu > -1$ and in-group bias is present in the population ($\bar{\pi} > 0$).*

2.2.1 The Role of Randomness and In-Group Reinforcement

Let $s^*(b_i, b_j) \equiv s(e_i^*(b_i, b_j), e_j^*(b_i, b_j))$ denote the aggregate effort in the NE of the social interaction stage game. Now, (3) can be written as a function only of biases b_i, b_j :

$$\rho_{b_i, b_j} = s^*(b_i, b_j) + (1 - s^*(b_i, b_j))[\beta(b_i, b_j) + (1 - \beta(b_i, b_j))(1 - \theta(\bar{\pi}))].$$

For a randomly drawn type τ individual, the expected probability of a positive interaction is

$$\bar{\rho}_\tau(\pi) = (1 - \pi_\tau)[(1 - \pi_{-\tau})\rho_{0,0} + \pi_{-\tau}\rho_{0,1}] + \pi_\tau[(1 - \pi_{-\tau})\rho_{1,0} + \pi_{-\tau}\rho_{1,1}],$$

which implies that

$$\pi_\tau^{t+1} = (1 - \eta_\tau)(1 - \bar{\rho}_\tau(\pi^t)) + \eta_\tau(\pi_\tau^t + (1 - \pi_\tau^t)\varepsilon)$$

Hence,

$$\Delta\pi_\tau = (1 - \eta_\tau) [1 - \bar{\rho}_\tau(\pi) - \pi_\tau] + \eta_\tau(1 - \pi_\tau)\varepsilon. \quad (6)$$

The role of randomness ($\varepsilon > 0$) is immediately apparent. It is necessary for out-group contact to affect steady-state bias in the population. When $\varepsilon = 0$, out-group interaction, and thus changes in the frequency of out-group contact, affect only the speed of adjustment, not the long-run level of bias.

I interpret $\eta_\tau(1 - \pi_\tau)\varepsilon$ as a reduced-form socialization or identity-reinforcement mechanism. Conditional on in-group interaction, a fraction ε of initially unbiased individuals adopt biased beliefs or norms through exposure to in-group narratives, conformity incentives, and local transmission. This interpretation is consistent with economic models of cultural transmission in which in-group socialization and assortative matching yield persistent group-specific attitudes in the population (Bisin and Verdier, 2000, 2001; Carvalho and Sacks, 2024; Carvalho et al., 2024a).¹³ Identity-based frameworks similarly imply that social categories and moral identity shape behavior and beliefs, and that in-group interactions can sustain and reinforce group-consistent worldviews (Akerlof and Kranton, 2000; Bénabou and Tirole, 2011). Because homophily and segregation raise the frequency of in-group interactions, even small in-group reinforcement rates can have quantitatively important implications for long-run bias (Currarini et al., 2009; Gentzkow and Shapiro, 2011).

3 Mechanism Decomposition

Before analyzing the full model, I decompose it into its three constituent mechanisms: effort, individual feedback, and societal feedback, and study the long-run behavior that each generates in isolation. This decomposition clarifies which aspects of the dynamics operate through each channel.

3.1 Effort

I begin with the effort channel in isolation. Can effort alone generate persistent effects under increased out-group contact? It cannot. Although effort responds to and shapes the probability of a positive interaction, effort admits a unique globally stable steady state. Therefore, temporary changes in μ or V are only transitory. Once μ or V reverts, so too does the steady state.

Fix $\beta(b_i, b_j) = 0$ and $\theta(\bar{\pi}) = 1$ for all b_i , b_j , and $\bar{\pi}$. In this case, the probability of a positive

¹³See Shayo (2020) for a survey of empirical and experimental evidence on in-group bias and in-group conformity.

interaction is $\rho_{b_i, b_j}(e_i, e_j) = s(e_i, e_j)$. By (5),

$$e_i^* = \frac{V}{1+V} \equiv e^*.$$

Evaluating the probability of a positive interaction at (e^*, e^*) yields

$$s(e^*, e^*) = \frac{V(2+V)}{(1+V)^2} \equiv s^* \in (0, 1).$$

Thus, when $V > \sqrt{2} - 1$, a positive interaction more likely than a negative one and vice versa.

The evolution of each π_τ is given by

$$\Delta\pi_\tau = (1 - \eta_\tau) [(1 - \pi_\tau)(1 - s^*) - \pi_\tau s^*] + \eta_\tau(1 - \pi_\tau)\varepsilon, \quad (7)$$

which yields the unique steady-state values

$$\pi_\tau^\infty = \begin{cases} 1 - \frac{(1-\eta_\tau)s^*}{1-\eta_\tau(1-\varepsilon)} & \text{if } \varepsilon > 0 \text{ or } \mu > -1 \\ \pi_\tau^0 & \text{if } \varepsilon = 0 \text{ and } \mu = -1. \end{cases} \quad (8)$$

If $\varepsilon = 0$ and $\eta_\tau = 1$ (i.e., $\mu = -1$), matching is purely assortative and biases remain constant. Otherwise, $\Delta\pi_\tau$ is linear and strictly decreasing in π_τ with $\Delta\pi_\tau(0) > 0 > \Delta\pi_\tau(1)$, so the dynamics converge globally to (8) from all initial conditions. The steady state is strictly decreasing in both μ (when $\varepsilon > 0$) and V (when $\mu > -1$), as can be verified by differentiating (8). $\varepsilon > 0$ and $\mu > -1$, the majority is strictly more biased in steady state than the minority ($\pi_1^\infty > \pi_2^\infty$ because $\eta_1 > \eta_2$) implies the majority accumulates more in-group reinforcement.

The effort benchmark delivers a clean comparative static. When $\varepsilon > 0$, increasing out-group contact (higher μ) strictly reduces steady-state bias. Increased contact reduces the weight on in-group interactions and increases the frequency of out-group encounters in which costly effort can generate positive interaction. The effect is amplified by V , as higher-value interactions increase effort and the corresponding probability of success.

This mechanism aligns with the canonical contact hypothesis, which emphasizes the role of structured, high-stakes cooperation in improving out-group attitudes (Allport, 1954; Pettigrew and Tropp, 2006), and with field evidence showing that sustained collaborative exposure reduces prejudice and discrimination (e.g., Lowe, 2021; Scacco and Warren, 2018; Bazzi et al., 2019). A temporary increase in contact shifts the transition path but not the steady state distribution. Once μ returns to its original value, so too does the steady state. The same logic holds for V . This is consistent

with evidence that the effects of short-lived contact interventions attenuate once contact ends. Absent continued interaction, the effects frequently decay over time (Paluck et al., 2021; Macchiavello and Morjaria, 2015). Permanent effects, by contrast, tend to arise in settings where out-group interaction is sustained or integrated into institutions (Bazzi et al., 2019).

3.2 Individual Feedback

I next isolate the individual-bias channel by shutting down effort and societal feedback. Unlike the effort-only benchmark, the model can now admit biased, unbiased, or interior steady states depending on parameters. However, for any fixed parameter configuration and $\mu > -1$, the dynamics converge to a unique steady state. Like the effort benchmark, temporary changes in out-group contact therefore affect only the transition path, not the steady state biases. V has no effect, as it only operates through effort.

Fix $s(e_i, e_j) = 0$ and $\theta(\bar{\pi}) = 1$ for all e_i, e_j , and $\bar{\pi}$. In this case, the probability of a positive interaction is given by $\rho_{b_i, b_j}(e_i, e_j) = \beta(b_i, b_j)$ and the evolution of π_τ is given by

$$\Delta\pi_\tau = (1 - \eta_\tau) \left[(1 - \pi_\tau) \left((1 - \pi_{-\tau})(1 - \beta_0) + \pi_{-\tau}(1 - \beta_1) \right) - \pi_\tau(1 - \pi_{-\tau})\beta_1 \right] + \eta_\tau(1 - \pi_\tau)\varepsilon. \quad (9)$$

Setting $\Delta\pi_\tau = 0$ and solving for π_τ yields the steady-state mapping

$$F_\tau(\pi_{-\tau}; \varepsilon) = \frac{(1 - \eta_\tau)[(1 - \pi_{-\tau})(1 - \beta_0) + \pi_{-\tau}(1 - \beta_1)] + \eta_\tau\varepsilon}{(1 - \eta_\tau)[(1 - \pi_{-\tau})(1 - \beta_0 + \beta_1) + \pi_{-\tau}(1 - \beta_1)] + \eta_\tau\varepsilon}, \quad (10)$$

so

$$\frac{\partial F_\tau(\pi_{-\tau}; \varepsilon)}{\partial \pi_{-\tau}} = \frac{(1 - \eta_\tau)\beta_1[(1 - \eta_\tau)(1 - \beta_1) + \eta_\tau\varepsilon]}{((1 - \eta_\tau)[(1 - \pi_{-\tau})(1 - \beta_0 + \beta_1) + \pi_{-\tau}(1 - \beta_1)] + \eta_\tau\varepsilon)^2}. \quad (11)$$

The biased steady state always exists as $\rho_{1,1}(0, 0) = 0$ implies no endogenous force reduces bias when it is universal. When $\varepsilon = 0$ and $\mu = -1$, steady states are symmetric and determined by the β_0/β_1 partition. If $\beta_1 > \frac{1}{2}$ and $\beta_0 < 1$, then there is a unique interior steady state

$$\pi_1^\infty = \pi_2^\infty = \frac{1 - \beta_0}{2\beta_1 - \beta_0}, \quad (12)$$

to which almost all initial conditions converge. When $\beta_1 \leq \frac{1}{2}$, out-group interactions are insufficiently bias-reducing and the biased steady state is globally attracting. Introducing $\varepsilon > 0$ eliminates the unbiased steady state and shifts any interior steady state upward. Full bias becomes globally stable whenever out-group updating near $\pi = (1, 1)$ is locally too weak to offset in-group reinforcement. The full characterization is in the Appendix.

Except under full assortativity ($\mu = -1$), steady states are independent of μ . Varying μ affects the speed of convergence but not the steady state. When $\beta_1 \leq \frac{1}{2}$, out-group interactions either reduce bias only weakly or reinforce it, and the population converges to full bias. In this region, increasing out-group contact increases the speed at which bias accumulates rather than mitigating it. This feature is consistent with empirical evidence showing that contact reduces prejudice primarily when interactions are cooperative and generate positive informational signals, and may instead entrench bias when interactions are conflict-oriented or unstructured (Allport, 1954; Pettigrew and Tropp, 2006; Lowe, 2021; Enos, 2014).

When $\varepsilon > 0$, the dynamics incorporate in-group reinforcement in addition to out-group updating. Even when out-group interactions are mildly positive, in-group contact generates drift toward bias. The unbiased steady state therefore disappears except under full assortativity, and any interior steady state shifts upward. Long-run outcomes are determined by the balance between positive bias-reducing out-group interactions and bias reinforcement near the fully biased state.¹⁴ Full bias becomes globally stable whenever out-group updating is locally too weak to offset in-group reinforcement.

This mechanism resonates with models of cultural transmission and identity formation in which in-group socialization sustains persistent group-specific beliefs (Bisin and Verdier, 2000, 2001; Akerlof and Kranton, 2000; Bénabou and Tirole, 2011; Carvalho and Sacks, 2024). Recent empirical evidence further suggests that contact alone is insufficient to reduce polarization when informational environments amplify in-group narratives. Studies of media consumption and online platforms document how selective exposure and social reinforcement can entrench partisan attitudes even amid out-group interaction (Gentzkow and Shapiro, 2011; Allcott and Gentzkow, 2017; Boxell et al., 2017). Experimental evidence likewise shows that out-group interactions may have limited or even backfiring effects when interactions trigger defensive identity responses (Bail et al., 2018).¹⁵

Because the in-group interaction rates η_1 and η_2 generally differ (except under full assortativity, $\mu = -1$), the interior steady state typically features asymmetric bias across types when $\varepsilon > 0$. This raises a natural comparative question: when there is an interior steady state, is the majority more or less biased than the minority in the long run?

Corollary 2. *Suppose $\mu > -1$, $\varepsilon > 0$ and an interior steady state exists. Then, $\pi_1^\infty > \pi_2^\infty$.*

¹⁴The full characterization of this trade-off is in Proposition A1 in the Appendix.

¹⁵Additional theoretical evidence includes Carvalho, Koyama and Sacks (2017) and Carvalho et al. (2024b).

For any $\mu > -1$, the majority's in-group interaction rate exceeds that of the minority's: $\eta_1 - \eta_2 = (2q - 1)(1 + \mu) > 0$, so the majority experiences less out-group contact than the minority. When $\varepsilon > 0$, within-type interactions generate drift toward bias, so a higher η_τ shifts type τ 's steady-state response upward. As a consequence, whenever the limiting state is interior, the majority ends up strictly more biased than the minority.

Relatedly, when $\varepsilon > 0$, increasing μ generally lowers steady-state bias. Because in-group reinforcement operates through η_τ , a rise in μ weakens the drift toward bias and strengthens the relative force of out-group contact-based bias reduction. As a result, parameters for which the biased steady state is locally stable can cross into a region in which an interior steady state exists. However, this change is not path-dependent. The steady states are pinned down by the contemporaneous value of μ . If μ reverts to its original level and the parameter configuration again implies that the biased steady state is the unique steady state, the dynamics converge back to that steady state.

The absence of path dependence in the individual-bias channel helps rationalize a recurring empirical finding. Many contact interventions generate short-run improvements in attitudes that dissipate once the intervention ends (Scacco and Warren, 2018; Paluck et al., 2019, 2021). Temporary increases in μ alter the transition path but not the steady state unless they permanently change reinforcement. Bias reverts once interaction patterns return to baseline. Durable effects arise only when contact reshapes the underlying reinforcement environment itself.

3.3 Societal Feedback

I now isolate the societal-feedback channel by shutting down effort and individual feedback. Can aggregate social feedback alone can generate persistent effects from increases in out-group contact? The answer now depends on the shape of $\theta(\bar{\pi})$. When $\theta(\bar{\pi})$ is linear or convex, the dynamics do not generate threshold-based path dependence from temporary changes in out-group contact. When $\theta(\bar{\pi})$ is sufficiently sigmoidal, threshold effects can emerge, giving rise to multiple steady states and distinct basins of attraction.

Fix $\beta(b_i, b_j) = s(e_i, e_j) = 0$ for all e_i, e_j, b_i and b_j . In this case, the probability of a positive interaction is given by $\rho_{b_i, b_j}(e_i, e_j) = 1 - \theta(\bar{\pi})$, yielding the evolutionary dynamic

$$\Delta\pi_\tau = (1 - \eta_\tau)(\theta(\bar{\pi}) - \pi_\tau) + \eta_\tau(1 - \pi_\tau)\varepsilon. \quad (13)$$

Hence, steady states are given by the solutions to the system

$$\pi_\tau = \gamma(\eta_\tau, \varepsilon)\theta(\bar{\pi}) + [1 - \gamma(\eta_\tau, \varepsilon)] \times 1, \quad (14)$$

where

$$\gamma(\eta_\tau, \varepsilon) = \frac{1 - \eta_\tau}{1 - \eta_\tau(1 - \varepsilon)} \equiv \gamma_\tau \in (0, 1),$$

which is strictly decreasing in both η_τ and ε (and thus increasing in μ). Define $\bar{\gamma} = q\gamma_1 + (1 - q)\gamma_2$.

This system can be reduced by studying the aggregate prevailing bias $\bar{\pi}$. Using (14) to compute $\bar{\pi} = q\pi_1 + (1 - q)\pi_2$ yields

$$\bar{\pi} = \bar{\gamma}(\theta(\bar{\pi}) - 1) + 1. \quad (15)$$

Thus, the steady states directly depend on the shape of $\theta(\bar{\pi})$. I consider three possibilities for θ : linear, convex, and sigmoidal.¹⁶ When linear, $\theta(\bar{\pi}) = \bar{\pi}$. When sigmoidal, I assume

$$\lim_{\bar{\pi} \rightarrow 0^+} \frac{d\theta(\bar{\pi})}{d\bar{\pi}} < 1, \quad \lim_{\bar{\pi} \rightarrow 1^-} \frac{d\theta(\bar{\pi})}{d\bar{\pi}} < 1.$$

These assumptions are consistent with Section 2.1 for $k = 1$ (linear) and k large (sigmoidal).

Thus, the steady states directly depend on the shape of $\theta(\bar{\pi})$. When θ is linear or convex, the dynamics do not generate threshold-based path dependence from temporary changes in out-group contact; steady states are either unique or, in the knife-edge linear case, pinned down by initial conditions.¹⁷ The novelty arises when feedback is sufficiently sigmoidal. In that case, small differences in aggregate bias can be amplified rather than dampened, generating tipping points and multiple steady states.¹⁸

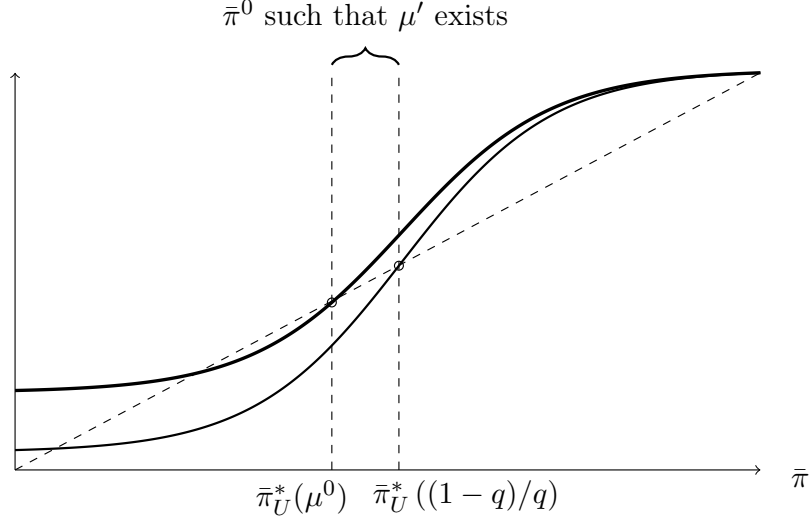
Proposition 1. *Fix $s(e_i, e_j) = 0$ and $\beta(b_i, b_j) = 0$ for all e_i, e_j, b_i , and b_j . Suppose $\varepsilon > 0$ and $\mu > -1$.*

- (i) *If $\theta(\bar{\pi})$ is sigmoidal and $\frac{d\theta(\bar{x})}{d\bar{\pi}} > \bar{\gamma}^{-1}$ for some $x \in (0, 1)$, then there are three steady states: a stable low-bias interior steady state $\bar{\pi}^*$, an unstable tipping point $\bar{\pi}_U^* > \bar{\pi}^*$, and the biased steady state at $(1, 1)$.*

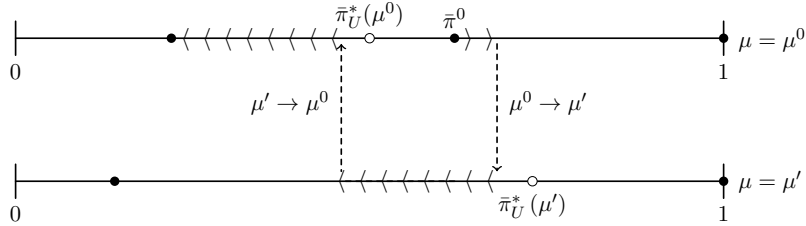
¹⁶The community norm enforcement foundation in Section 2.1 provides a concrete case, $k = 1$, $k = 2$, and $k \geq 3$, respectively, for these three shapes.

¹⁷The full characterization is in Proposition A1 in the Appendix.

¹⁸Allowing θ to be concave does not substantially add anything to the results. When $\varepsilon = 0$, there is convergence to the biased steady state from almost all initial conditions and when $\varepsilon > 0$, the results are identical to the linear case. See the proof for details



(a) Evolution of $\bar{\pi}$ under $\mu = \mu^0$ and $\mu = \frac{1-q}{q}$.



(b) Transition path given initial biases (π_1^0, π_2^0) and initial average bias $\bar{\pi}^0 = q\pi_1^0 + (1-q)\pi_2^0$ and temporary increase in μ from μ^0 to μ' .

Figure 2: Graphical illustration of Proposition 1.

- (ii) Under the conditions of (i), if $\bar{\pi}^0 \in (\bar{\pi}_U^*(\mu^0), \bar{\pi}_U^*(\frac{1-q}{q}))$, then there exists a $\mu' > \mu^0$ such that temporarily increasing μ^0 to μ' yields a persistent reduction in aggregate bias.

The proposition delivers the minimal mechanism through which temporary contact can generate persistent change. Sigmoidal societal feedback establishes a tipping point, and contact need only push society across it. The intervention need not be permanent; it need only last long enough to move the economy across the tipping point into the low bias basin of attraction. This rationalizes why some contact interventions produce durable change while others do not. Contact has lasting effects only when it shifts the society across the relevant tipping point. Figure 2 illustrates this mechanism.

The analysis thus far isolates the minimal mechanism required for temporary contact to generate permanent change. In practice, however, effort, individual feedback, and societal feedback operate

simultaneously. I now reintroduce all three channels and characterize the full dynamics of the model, showing how their interaction determines when contact reduces bias and when it instead reinforces high-bias outcomes.

4 Dynamics of the Full Model

I now reintroduce all three channels: effort, individual feedback, and societal feedback. The full model nests each benchmark in Section 3, but their interaction generates new forces absent in isolation. Effort counteracts in-group bias. As a result, the evolution of bias may be either globally convergent to a unique steady state or threshold-driven, depending on parameters. This section characterizes steady states of the full system and identifies when temporary increases in out-group contact can generate permanent change.

Lemma 2. *Conditional on individuals i and j matching with biases b_i and b_j , the probability of a positive interaction is monotonic in $\bar{\pi}$ iff*

$$V \leq \hat{V}(b_i, b_j) \equiv \frac{1}{1 - \beta(b_i, b_j)}.$$

Lemma 2 highlights the key interaction between effort, individual feedback, and societal feedback. In the societal-feedback benchmark of Section 3.3, higher aggregate bias mechanically reduced the probability of a positive interaction. Incorporating effort, individuals respond to this bias. When V is small, effort is weak and the probability of success declines monotonically in $\bar{\pi}$. When V is sufficiently large, effort responds strongly enough that increases in $\bar{\pi}$ induce additional investment, dampening the effect of aggregate bias on interaction outcomes and potentially rendering success probabilities non-monotonic in $\bar{\pi}$. High-value environments therefore attenuate the propagation of bias. This has a direct policy implication. V -based interventions may be most effective precisely in highly biased societies, where the incentive to invest in effort is greatest. This reverses the intuition from the benchmark, where high-bias environments are exactly the ones most resistant to intervention.

Effort also eliminates the fully biased steady state.

Lemma 3. *Suppose that $\mu > -1$. The biased steady state does not exist. Additionally, the unbiased steady state exists iff $\varepsilon = 0$.*

In the individual feedback and societal feedback benchmarks, the fully biased steady state was self-sustaining because no endogenous force reduced bias once it was universal. With effort, however, biased individuals still invest to secure positive interactions. As long as $\mu > -1$, out-group contact occurs with positive probability, and effort generates positive interactions that reduce bias. The biased steady state therefore disappears.

To characterize the multiplicity of steady states, define the *slope envelope* as

$$\tilde{p}'(\bar{\pi}) = \frac{d\theta(\bar{\pi})}{d\bar{\pi}} \times \max_{\beta \in [0,1]} \frac{(1-\beta)|1 - (1-\beta)\theta(\bar{\pi})V|}{(1 + (1-\beta)\theta(\bar{\pi})V)^3},$$

and set $\tilde{P}' := \sup_{\bar{\pi} \in [0,1]} \tilde{p}'(\bar{\pi})$, which characterizes the maximal steepness of the inflection point when societal feedback is self-reinforcing (sigmoidal). Additionally, define

$$\bar{\Delta} = \sup_{\bar{\pi} \in [0,1]} \max_{\beta, \beta' \in \{0, \beta_1, \beta_0\}} |m_\beta(\bar{\pi}) - m_{\beta'}(\bar{\pi})| \geq 0,$$

where $m_\beta(\bar{\pi}) = 1 - \rho_{b_i, b_j}$ for $\beta = \beta(b_i, b_j)$.

The term $\bar{\Delta}$ is the largest gap in interaction failure probabilities across all levels of aggregate bias and bias configurations.

Proposition 2. *Suppose $\mu > -1$. The following statements hold.*

- (i) *If $2\bar{\Delta} + \tilde{P}' < \gamma_2^{-1}$, then there is a unique steady state $(\pi_1^\infty, \pi_2^\infty) \in [0, 1]^2$.*
- (ii) *Suppose $\varepsilon > 0$ and $\frac{dm_{\beta_0}(\bar{\pi})}{d\bar{\pi}}$ is single-peaked on $(0, 1)$ with $\bar{\pi}_M = \arg \max_{\bar{\pi}} \frac{dm_{\beta_0}(\bar{\pi})}{d\bar{\pi}} \in (0, 1)$. There exists a $\hat{\Delta}$ such that if $\tilde{p}'(0), \tilde{p}'(1) < \bar{\gamma}^{-1} < \frac{dm_{\beta_0}(\bar{\pi}_M)}{d\bar{\pi}}$ and $\bar{\Delta} < \hat{\Delta}$, then there are at least three interior steady states.*

Let $\bar{\pi}_1^*$, $\bar{\pi}_2^*$, and $\bar{\pi}_3^*$ denote the steady state aggregate bias levels. Here, $\bar{\pi}_2^*$ is the tipping point, analogous to $\bar{\pi}_U^*$ from Section 3.3. Because $\tilde{P}'$ is decreasing in V , uniqueness in statement (i) is more easily satisfied when V is large, individuals then respond aggressively to rising bias, dampening the aggregate feedback required for multiplicity. Effort therefore attenuates the nonlinearities necessary for tipping dynamics. Invoking the community norm enforcement interpretation of $\theta(\bar{\pi})$ from Section 2.1, for any $V > 0$ and maintaining $\varepsilon > 0$ and $\bar{\Delta} < \hat{\Delta}$, there always exists k sufficiently large to satisfy the slope conditions of (ii). The latter condition is least restrictive precisely when tipping dynamics are most pronounced (see footnote 28 in the Appendix).

Let $\Omega = (\theta, \beta_0, V, q, \mu, \varepsilon)$ denote the set of parameters under which the full model admits three steady states and let Ω_S denote the analogous set under the societal feedback benchmark. Similarly, let $\tilde{\Omega}$ and $\tilde{\Omega}_S$ denote the respective sets such that a temporary increase in μ can generate a persistent reduction in long run bias under the full model and societal-feedback benchmark.

The basin-switching logic of Proposition 1 extends to the full model. Under the conditions of Proposition 2, if $\bar{\pi}^0 \in (\bar{\pi}_2^*(\mu^0), \bar{\pi}_2^*(\frac{1-q}{q}))$, then there exists a $\mu' > \mu^0$ such that temporarily increasing μ^0 to μ' yields a persistent reduction in steady-state bias with convergence to $\bar{\pi}_1^*(\mu^0)$.¹⁹

However, because individuals increase effort when $\bar{\pi}$ rises, the feedback from aggregate bias to contact outcomes is dampened, and the parameter region supporting multiplicity shrinks. The following proposition shows that this shrinkage is monotone in V . As the value of successful interaction rises, the set of environments admitting μ -based tipping contracts strictly.

Proposition 3. *Suppose $\mu > -1$, $\varepsilon > 0$, and $\beta_0 \in [0, 1)$. For each $\bar{\pi} > 0$, the pointwise slope envelope $\tilde{p}'(\bar{\pi}; V)$ is strictly decreasing in V . Consequently:*

- (i) $\tilde{P}'(V) = \sup_{\bar{\pi} \in [0, 1]} \tilde{p}'(\bar{\pi}; V)$ and $\bar{\Delta}(V)$ are both strictly decreasing and continuous in V .
- (ii) For any $V' > V > 0$, $\Omega(V') \subset \Omega(V)$. That is, the parameter region supporting multiple steady states shrinks strictly as V rises.

Proposition 3 delivers a monotone comparative static that strengthens the cross-model comparison with the societal-feedback benchmark. Setting $V = 0$ recovers the benchmark (effort is absent and $m_{\beta_0}(\bar{\pi}; 0) = (1 - \beta_0)\theta(\bar{\pi})$), so $\Omega(V) \subset \Omega_S$ for any $V > 0$ with strict inclusion when $\beta_0 \in [0, 1)$. The within-model result goes further. As V rises from any baseline, the multiplicity region contracts strictly, and the basin-switching region $\tilde{\Omega}$ contracts with it.

Taken together, the full model reveals a central tradeoff introduced by effort. Higher-value contact reduces steady-state in-group bias by inducing preventive investment in effort, thereby stabilizing the dynamics. At the same time, this stabilization weakens the sensitivity of contact outcomes to aggregate bias. Because individuals respond to rising $\bar{\pi}$ by increasing effort, the societal feedback loop is dampened and the nonlinearity required for tipping dynamics is harder to sustain. High-value contact environments therefore lower steady-state in-group bias while simultaneously limiting the scope for temporary exposure to generate permanent regime shifts. In this sense, effort changes

¹⁹The proof, which follows the benchmark argument with Ψ replacing H , is in the Appendix.

both the level and the shape of the dynamics. It lowers long-run bias, but it also shrinks the set of environments in which μ -based policy succeeds.

If the economy is deeply entrenched in high bias, so that $\bar{\pi}^0$ lies above the window $(\bar{\pi}_2^*(\mu^0), \bar{\pi}_2^*(\frac{1-q}{q}))$, it cannot be rescued by any temporary increase in μ . The following proposition shows that, under the maintained assumptions, increasing V provides an alternative route to permanent change that does not require the initial condition to lie inside the μ -window. Let $\bar{\pi}_\ell^*(V)$ denote the steady states as functions of V .

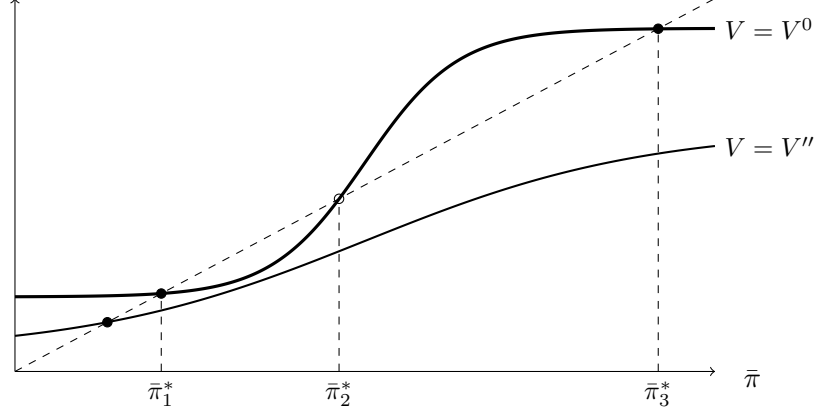
Proposition 4. *Suppose $\varepsilon > 0$, $\mu > -1$, and parameters are contained in Ω .*

- (i) *There exists $V' > V^0$ such that for all $V \geq V'$, the full model admits a unique globally attracting steady state $\bar{\pi}_1^*(V) \in (0, 1)$, with $\bar{\pi}_1^*(V) < \bar{\pi}_2^*(V^0)$.*
- (ii) *For any $\bar{\pi}^0 \in (\bar{\pi}_2^*(V^0), 1)$, temporarily increasing V from V^0 to any $V'' > V'$ satisfying (i) yields a persistent reduction in long-run bias, with the economy converging to $\bar{\pi}_1^*(V^0)$.*

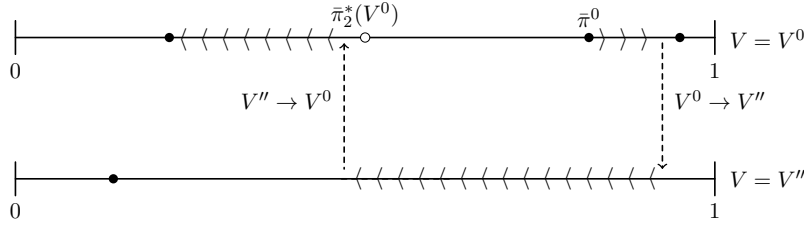
Proposition 4 complements the μ -intervention but works through a fundamentally different mechanism.²⁰ A temporary increase in μ works by shifting the tipping point $\bar{\pi}_2^*(\mu)$ upward, expanding the basin of attraction of the low-bias steady state until the current state falls inside it; this requires $\bar{\pi}^0$ to lie within a specific window of initial conditions. A temporary increase in V works by raising the return to effort. Individuals then invest heavily in positive contact even in high-bias environments, eroding the self-reinforcing dynamic that sustains the high-bias trap and leaving a unique low-bias steady state. During the intervention, aggregate bias reduces towards the unique low-bias steady state. The restriction on initial conditions near the tipping point disappears entirely. Figure 3 illustrates this mechanism.

The two instruments therefore differ in robustness. μ -based interventions are sufficient only when the economy is already near the tipping point, whereas V -based interventions admit no such constraint. Proposition 4 shows that a sufficiently strong V -intervention generates permanent change from any initial condition (provided that aggregate bias was trending towards the high-bias steady state), while Proposition A3 shows that a μ -intervention succeeds only when prevailing bias lies within a window around the tipping point. This asymmetry implies that the two instruments are complements rather than substitutes. A temporary V -intervention can move a deeply entrenched

²⁰Proposition A3 in the Appendix establishes the existence of the μ -intervention in the full model.



(a) Evolution of $\bar{\pi}$ under $V = V^0$ and $V = V''$.



(b) Transition path given initial biases (π_1^0, π_2^0) and initial average bias $\bar{\pi}^0 = q\pi_1^0 + (1-q)\pi_2^0$ and temporary increase in V from V^0 to V'' .

Figure 3: Graphical illustration of Proposition 4.

economy into the window from which a subsequent μ -intervention completes the transition to the low-bias steady state, allowing the V -intervention to be withdrawn earlier than under a pure V -policy. Formalizing this sequencing logic requires two preliminary results.

The first derives a bound on how fast aggregate bias can fall in a single period under the baseline matching structure.

Lemma 4. *For any $\mu > -1$, $\bar{\pi}^t - \bar{\pi}^{t+1} \leq 2q(1-q)(1+\mu) \equiv C^*(\mu, q)$ for all $t \geq 0$.*

Second, the unstable tipping point moves monotonically in both policy instruments. Under Ω with $\mu > -1$ and $\varepsilon > 0$, the unstable tipping point $\bar{\pi}_2^*(\mu, V)$ is strictly increasing in both μ and V .²¹

Fix a candidate sequential matching level $\tilde{\mu} \in (\mu^0, \frac{1-q}{q}]$ and a V -intervention level V'' satisfying

²¹This follows from the implicit function theorem. See the Appendix for a proof.

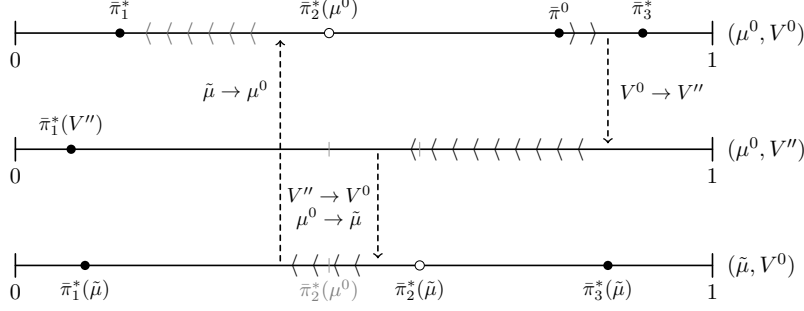


Figure 4: Transition path given initial biases (π_1^0, π_2^0) and initial average bias $\bar{\pi}^0 = q\pi_1^0 + (1-q)\pi_2^0$ and temporary increase in V from V^0 to V'' , followed by returning V'' to V^0 and instituting a temporary increase in μ from μ^0 to $\tilde{\mu}$.

Proposition 4(i). Define crossing times along the trajectory induced by the constant policy (V'', μ^0) :

$$T_V \equiv \min \{t \geq 0 : \bar{\pi}^t < \bar{\pi}_2^*(\mu^0, V^0)\},$$

$$\tilde{T}_V \equiv \min \{t \geq 0 : \bar{\pi}^t < \bar{\pi}_2^*(\tilde{\mu}, V^0)\},$$

and the μ -phase duration along the trajectory induced by $(\tilde{\mu}, V^0)$ initiated at $\bar{\pi}^{\tilde{T}_V}$:

$$T_\mu \equiv \min \{t \geq 0 : \bar{\pi}^{\tilde{T}_V+t} < \bar{\pi}_2^*(\mu^0, V^0)\}.$$

To obtain a nontrivial sequencing result, assume that a μ -intervention alone fails; i.e., for every feasible $\mu' \in (\mu^0, \frac{1-q}{q}]$, $\bar{\pi}^0 \geq \bar{\pi}_2^*(\mu', V^0)$. Additionally, assume that there exists $\tilde{\mu} \in (\mu^0, \frac{1-q}{q}]$ such that $C^*(\mu^0, q) < \bar{\pi}_2^*(\tilde{\mu}, V^0) - \bar{\pi}_2^*(\mu^0, V^0)$.

Proposition 5. *Suppose $\varepsilon > 0$, $\mu^0 > -1$, and parameters are contained in Ω for all $\mu \in [\mu^0, \frac{1-q}{q}]$, with stable aggregate bias levels $\bar{\pi}_1^*(\mu, V) < \bar{\pi}_3^*(\mu, V)$ and unstable tipping point $\bar{\pi}_2^*(\mu, V)$. Fix (μ^0, V^0) and $\bar{\pi}^0 \in (\bar{\pi}_2^*(\mu^0, V^0), 1)$. Then $T_V, \tilde{T}_V, T_\mu < \infty$ and the combined policy that raises V^0 to V'' on $t \in [0, \tilde{T}_V)$, μ^0 to $\tilde{\mu}$ on $t \in [\tilde{T}_V, \tilde{T}_V + T_\mu)$, then ends the intervention yields a persistent reduction in steady state aggregate bias. Additionally, $\tilde{T}_V < T_V$, so the V -intervention is withdrawn strictly earlier under sequencing than under the V -only policy.*

Figure 4 illustrates the transition path of the sequenced policy. Proposition 5 shows that sequencing can shorten the required V -phase, though not necessarily the total intervention duration. A μ -intervention need only last long enough to push the trajectory below the baseline tipping point $\bar{\pi}_2^*(\mu^0, V^0)$, whereas a pure V -intervention must do so without the aid of a temporary increase in μ . Whether sequencing reduces total duration depends on the speed of the μ -induced dynamics relative

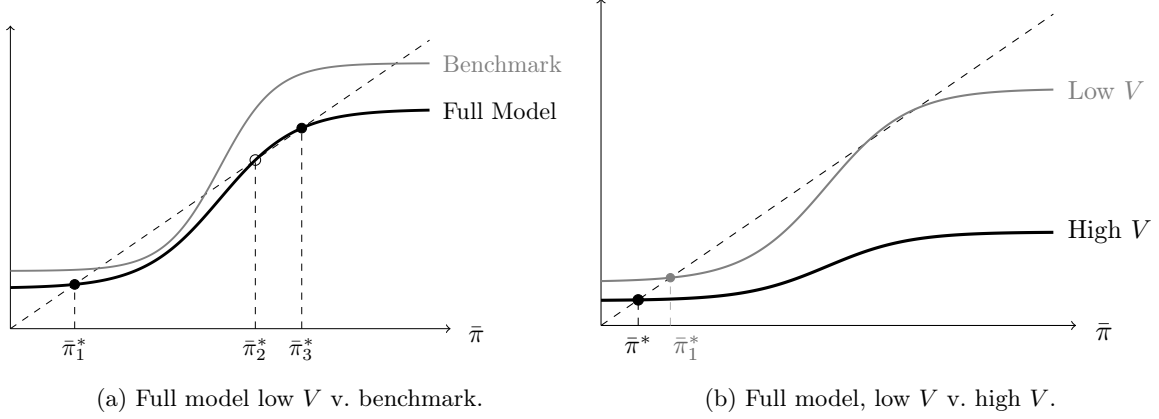


Figure 5: Endogenous effort attenuates the societal feedback sustaining tipping dynamics (Panel (a), Proposition 3). As V increases, this attenuation eliminates the high-bias steady state entirely (Panel (b), Proposition 4)

to the remaining V -phase $T_V - \tilde{T}_V$. A more aggressive $\tilde{\mu}$ widens the gap $\tilde{\pi}_2^*(\tilde{\mu}, V^0) - \tilde{\pi}_2^*(\mu^0, V^0)$, relaxing the assumption on $C^*(\mu^0, q)$ and shortening $\tilde{T}_V$, but may lengthen T_μ if the state enters the tipping window close to its upper boundary. The choice of $\tilde{\mu}$ therefore trades off the duration of the V -phase against that of the μ -phase.

The choice of $\tilde{\mu}$ involves a genuine tradeoff. $\tilde{\pi}_2^*(\tilde{\mu}, V^0)$ is strictly increasing in $\tilde{\mu}$, so a more aggressive matching mandate shortens the V -phase by raising the threshold the trajectory must cross before the μ -phase begins. However, its effect on the μ -phase duration T_μ is ambiguous. A higher $\tilde{\mu}$ speeds convergence once imposed, but it also begins from a state with higher prevailing bias. The preferred value of $\tilde{\mu}$ therefore need not equal the maximum feasible matching rate $(1 - q)/q$.

Sections 3 and 4 together clarify the conditions under which contact alters the steady-state in-group bias. When feedback is linear or convex, exposure affects long-run bias but does not generate threshold-driven regime shifts. Figure 5 illustrates the two central mechanisms. Endogenous effort shrinks the parameter region in which μ -based tipping is possible, but simultaneously creates the conditions under which V -based regime elimination becomes effective. The contact hypothesis therefore holds when social feedback is sigmoidal, and the instrument best suited to exploiting that feedback depends on where the population stands relative to the tipping point.

5 Welfare Effects of Contact- and Value-Interventions

The preceding section characterizes when each instrument can generate permanent regime change. This section asks which instrument does so at lowest welfare cost. The model's effort structure delivers a closed-form welfare function in which aggregate welfare equals the interaction prize minus a population-averaged effort tax that is increasing in aggregate bias. The low-bias steady state strictly welfare-dominates the high-bias steady state. V -interventions raise welfare on impact, while μ -interventions impose a front-loaded welfare penalty before eventually generating gains. Accounting for these transition-path effects can reverse the fiscal ranking of instruments.

Throughout, I maintain two assumptions: (i) the economy begins in the high-bias basin under (V^0, μ^0) , and (ii) $\bar{\pi}^0 \in (\bar{\pi}_2^*(\mu^0), \bar{\pi}_2^*(\frac{1-q}{q}))$, so that all three candidate policies, μ -only, V -only, and sequencing, are dynamically feasible and can be compared on equal footing. When (ii) fails, μ -only is ruled out by the dynamics of Section 4 and the comparison reduces to V -only versus sequencing.

5.1 The Welfare Function and Steady-State Ranking

Recall from (4) that i 's utility, when matched with j can be written as

$$u_i = \rho_{b_i, b_j} V - \frac{1}{2} e_i^2.$$

In equilibrium, the probability of success is $\rho = 1 - (1 - s^*)(1 - \beta)\theta$. Using $e^* = \frac{z}{1+z}$, where $z = (1 - \beta)\theta V$ (by Lemma 1), $\rho V = V - \frac{z}{(1+z)^2}$, and $\frac{(e^*)^2}{2} = \frac{z^2}{2(1+z)^2}$. Therefore,

$$u^* = V - \frac{z(2+z)}{2(1+z)^2} = V - \frac{1}{2} s^*,$$

as $s^* = f(z) = \frac{z(2+z)}{(1+z)^2}$ and thus $1 - s^* = \frac{1}{(1+z)^2}$. Every match yields the full prize V minus an “effort tax” of $\frac{s^*}{2}$ that accounts for environmental adversity induced by the tension between biases and a positive interaction.

For any state π , let s_β^* denote the equilibrium aggregate effort given the bias pair β_k , $k \in \{0, 1, 2\}$ and let

$$\begin{aligned} \bar{S}(\pi) &= [q\eta_1 + (1-q)\eta_2] s_0^* + [q(1-\eta_1) + (1-q)(1-\eta_2)] \\ &\quad \times \left[(1-\pi_1)(1-\pi_2) s_0^* + ((1-\pi_1)\pi_2 + \pi_1(1-\pi_2)) s_1^* + \pi_1\pi_2 s_2^* \right] \end{aligned}$$

denote the average aggregated effort of the population in the Nash equilibrium of the social interaction stage game. Then, the welfare function can be written as

$$W(\pi) = V - \frac{1}{2}\bar{S}(\pi). \quad (16)$$

This expression shows that welfare is determined entirely by V and population-average effort. In-group matches contribute s_0^* (effort against the no-bias environment), while out-group matches contribute a weighted average of s_0^* , s_1^* , and s_2^* , where the weights depend on the bias composition.

Observe from above that $f(z)$ is strictly increasing in z and that $\frac{df(z)}{dz} = \frac{2}{(1+z)^3} > 0$. As $z_k = (1 - \beta_k)\theta(\bar{\pi})V$ is increasing in $\bar{\pi}$ through $\theta(\bar{\pi})$, and the bias composition shifts toward higher- z match types as $\bar{\pi}$ rises, $\bar{S}$ is strictly increasing in $\bar{\pi}$ on any path where all three effects are aligned. It follows immediately that for any two interior steady states with $\bar{\pi}_1^* < \bar{\pi}_3^*$, the low-bias steady state strictly welfare-dominates the high-bias one: $W(\bar{\pi}_1^*) > W(\bar{\pi}_3^*)$. The ranking reflects two effects: a direct feedback effect and a composition effect. High-bias environments require more effort to sustain positive interactions ($\theta(\bar{\pi}_3^*) > \theta(\bar{\pi}_1^*)$) and, in the high-bias steady state, more individuals are biased, so a larger share of out-group matches are biased pairs (with $\beta = 0$) and a smaller share are unbiased pairs (with $\beta = \beta_0$). The distribution of matches shifts toward higher-effort types, lowering welfare. Define the welfare gap as

$$\Delta W = \frac{1}{2} [\bar{S}(\bar{\pi}_3^*) - \bar{S}(\bar{\pi}_1^*)]. \quad (17)$$

Now, suppose that all individuals discount future payoffs by the common factor $\delta \in (0, 1)$. The present discounted value of steady-state welfare is $\frac{W}{1-\delta}$, so the long-run value of transitioning permanently from $\bar{\pi}_3^*$ to $\bar{\pi}_1^*$ (the *value of regime change* R) is

$$R = \frac{\Delta W}{1-\delta} = \frac{\bar{S}(\bar{\pi}_3^*) - \bar{S}(\bar{\pi}_1^*)}{2(1-\delta)}. \quad (18)$$

R captures the present value of the perpetual per-period reduction in effort costs from operating in a low-bias environment.

5.2 Transition-Path Welfare

To incorporate the transition dynamics, I define the V -bonus $B_V(t)$, which captures the per-period welfare gains under a V -intervention. Let

$$B_V(t) = W(\bar{\pi}^t; V'', \mu^0) - W(\bar{\pi}^t; V^0, \mu^0) = (V'' - V^0) - \frac{1}{2} [\bar{S}(\bar{\pi}^t; V'', \mu^0) - \bar{S}(\bar{\pi}^t; V^0, \mu^0)]. \quad (19)$$

Observe that differentiating u^* with respect to V yields

$$\frac{\partial u^*}{\partial V} = 1 - \frac{(1 - \beta)\theta}{(1 + z)^3}$$

As $z \in (0, 1)$, $(1 - \beta)\theta \leq 1$ and $\frac{(1-z)}{(1+z)^3} < 1$, so $\frac{\partial u^*}{\partial V} > 0$. Hence, u^* is weakly increasing in V for every match type. Aggregating implies that $B_V(t) \geq 0$. Increasing V has two effects: it increases the prize from each successful interaction and it induces additional effort. The prize effect always dominates because the effort response is a second-order correction. Individuals optimally choose effort that equates the marginal cost to the marginal benefit, so the net payoff from additional effort is always non-negative by the envelope theorem. V -interventions are therefore welfare-improving in every period, even before bias adjusts.

Define $J_\mu(t)$ as the μ -penalty; i.e., the per-period welfare loss from operating under $\tilde{\mu}$ rather than μ^0 for a given π^t :

$$J_\mu(t) = W(\pi^t; V^0, \mu^0) - W(\pi^t; V^0, \tilde{\mu}) = \frac{1}{2} [\bar{S}(\pi^t; V^0, \tilde{\mu}) - \bar{S}(\pi^t; V^0, \mu^0)],$$

which can be rewritten as

$$J_\mu(t) = \frac{1}{2}(\tilde{\mu} - \mu^0)q(1 - q) \sum_{\tau} [\bar{s}_{out,\tau}^*(\pi^t) - s_0^*] \geq 0, \quad (20)$$

where $\bar{s}_{out,\tau}^*(\pi^t)$ is the average aggregate effort from a type- τ individual in an out-group match and s_0^* is the aggregate effort from an in-group match. The penalty is proportional to the mandate intensity $\tilde{\mu} - \mu^0$ and the effort-cost gap between out-group and in-group matches. The gap is largest when bias is high (many biased pairs with $\beta_2 = 0$) and shrinks as bias falls (more unbiased pairs with $\beta_0 > 0$, which require less effort). It's nonnegativity follows from the fact that effort is always at its lowest with either in-group matches or unbiased matches. Hence $\bar{s}_{out,\tau}^*(\pi^t) - s_0^* \geq 0$, with equality only if $\bar{\pi} = 0$, so $\pi^t = (1, 1)$. $J_\mu(t)$ is monotonically declining along any trajectory where bias is falling. As $\bar{\pi}$ falls, the fraction of high-effort biased out-group pairs shrinks, so s_{out} converges toward s_0^* , and the gap narrows.

Unlike V -interventions, which uniformly increase welfare (as $B_V(t)$ is nonnegative), μ -interventions exhibit a J -shape. Let $\bar{\pi}_{int}^t$ denote the aggregate bias at time t in the scenario in which there is a μ -intervention and let $\bar{\pi}_{cf}$ denote the counterfactual in which there is no intervention, with π_{int}^t and π_{cf}^t defined analogously. The net change in per-period welfare from the intervention is

$$W(\bar{\pi}_{int}^t; \tilde{\mu}) - W(\bar{\pi}_{cf}^t; \mu^0) = \frac{1}{2} [\bar{S}(\pi_{cf}^t; \mu^0) - \bar{S}(\pi_{int}^t; \tilde{\mu})].$$

Let $\bar{S}(\pi_{int}^t; \mu^0)$ denote the average aggregate effort if bias had followed the intervention trajectory but matching had stayed at baseline. Adding and subtracting this value from the change in welfare yields

$$\begin{aligned} W(\bar{\pi}_{int}^t; \tilde{\mu}) - W(\bar{\pi}_{cf}^t; \mu^0) &= \frac{1}{2} [\bar{S}(\pi_{cf}^t; \mu^0) - \bar{S}(\pi_{int}^t; \mu^0)] + \frac{1}{2} [\bar{S}(\pi_{int}^t; \mu^0) - \bar{S}(\pi_{int}^t; \tilde{\mu})] \\ &= \underbrace{\frac{1}{2} [\bar{S}(\pi_{cf}^t; \mu^0) - \bar{S}(\pi_{int}^t; \mu^0)]}_{\text{bias-reduction gain} \geq 0} + \underbrace{\frac{1}{2} [\bar{S}(\pi_{int}^t; \mu^0) - \bar{S}(\pi_{int}^t; \tilde{\mu})]}_{\mu\text{-penalty} \leq 0} = -J_\mu(t). \end{aligned}$$

At the start of the intervention t_1 , $\pi_{cf}^{t_s} = \pi_{int}^{t_1}$, so $W(\bar{\pi}_{int}^{t_1}; \tilde{\mu}) - W(\bar{\pi}_{cf}^{t_1}; \mu^0) < 0$ and the initial change in welfare is strictly negative. At some $\hat{t} > t_1$, $W(\bar{\pi}_{int}^{\hat{t}}; \tilde{\mu}) = W(\bar{\pi}_{cf}^{\hat{t}}; \mu^0)$ and remains positive for all t thereafter. As $t \rightarrow \infty$, the intervention converges to the low-bias steady state. The bias-reduction gain stabilizes at ΔW (the welfare gap). The penalty stabilizes at $J_\mu(\infty)$ evaluated at $\bar{\pi}_1$, which is small (bias is low and the effort gap is narrow). Total welfare gains are permanently positive.

Proposition 6. *Total welfare under a V -intervention relative to the counterfactual is strictly positive in every period. Total welfare under a μ -intervention relative to the counterfactual is initially negative, eventually positive during the intervention, and permanently positive thereafter.*

The μ -penalty $J_\mu(t)$ provides an economic foundation for political resistance to integration mandates. μ -interventions impose visible, immediate welfare costs on participants. They are being asked to engage in costlier out-group interactions while the bias-reduction gains materialize only later. V -interventions face no such asymmetry: $B_V(t) \geq 0$ from the start. This welfare-based rationale for V -first sequencing is distinct from the dynamic rationale in Proposition 5. V -first is preferred not only because it expands the basin of attraction, but also because it avoids front-loading welfare costs on participants.

5.3 Welfare-Efficient Intervention Design

Proposition 6 establishes that a transition from the high-bias to the low-bias steady state increases welfare, that V -interventions generate per-period welfare gains throughout the transition, and that μ -interventions impose front-loaded welfare costs that are eventually offset by bias-reduction gains. This subsection uses these properties to determine which instrument, or combination of instruments, achieves a successful reduction of in-group bias at the lowest welfare-adjusted cost.

Because all successful policies converge to the same low-bias steady state $\bar{\pi}_1^*(\mu^0, V^0)$, the long-run value of regime change R is identical across them. The welfare ranking therefore reduces to

a comparison of costs incurred during the intervention phase, where costs include both the fiscal expenditure on the policy and the transition-path welfare effects characterized above. To formalize this, define $c_V > 0$ and $c_\mu > 0$ denote the per-period (monetary) costs of the V - and μ -interventions, respectively. Then, define the *welfare-adjusted per-period costs*

$$\tilde{c}_V(t) \equiv c_V - B_V(t), \quad \tilde{c}_\mu(t) \equiv c_\mu + J_\mu(t). \quad (21)$$

The effective cost of a V -intervention in period t is its fiscal cost c_V minus the welfare bonus $B_V(t) \geq 0$ it generates. The effective cost of a μ -intervention in period t is its fiscal cost c_μ plus the welfare penalty $J_\mu(t) \geq 0$ it imposes. Accounting for transition-path welfare therefore makes V cheaper and μ more expensive than their fiscal costs alone suggest.

I compare three candidate policies, each of which successfully crosses the tipping point.

1. *Policy M (μ -only)*. Raise μ from μ^0 to μ' for T_M periods, where

$$T_M \equiv \min \{t \geq 0 : \bar{\pi}^t < \bar{\pi}_2^*(\mu^0, V^0)\}$$

along the trajectory induced by (μ', V^0) . This policy can only be successfully implemented when $\bar{\pi}^0 \in (\bar{\pi}_2^*(\mu^0, V^0), \bar{\pi}_2^*(\frac{1-q}{q}, V^0))$.

2. *Policy V (V -only)*. Raise V from V^0 to V'' for T_V periods. Feasible from any $\bar{\pi}^0 \in (\bar{\pi}_2^*(\mu^0, V^0), 1)$.

3. *Policy S (sequencing)*. Raise V from V^0 to V'' for $\tilde{T}_V$ periods, then raise μ from μ^0 to $\tilde{\mu}$ for T_μ periods. Feasible from any $\bar{\pi}^0 \in (\bar{\pi}_2^*(\mu^0, V^0), 1)$.

For each policy $P \in \{M, V, S\}$, the total welfare-adjusted cost is the discounted sum of welfare-adjusted per-period costs along its trajectory:

$$C_M = \sum_{t=0}^{T_M-1} \delta^t \tilde{c}_\mu^M(t), \quad (22)$$

$$C_V = \sum_{t=0}^{T_V-1} \delta^t \tilde{c}_V^V(t), \quad (23)$$

$$C_S = \sum_{t=0}^{\tilde{T}_V-1} \delta^t \tilde{c}_V^S(t) + \sum_{t=\tilde{T}_V}^{\tilde{T}_V+T_\mu-1} \delta^t \tilde{c}_\mu^S(t), \quad (24)$$

where the superscript on $\tilde{c}$ indicates that B_V and J_μ are evaluated along the respective policy's trajectory.

Proposition 7. *The welfare-optimal policy minimizes the total welfare-adjusted cost among $\{C_M, C_V, C_S\}$. The pairwise rankings are as follows.*

- (i) *V-only versus sequencing. V is welfare-preferred to S iff $C_V < C_S$.*
- (ii) *V-only versus μ -only. V is welfare-preferred to M iff $C_V < C_M$.*
- (iii) *Sequencing versus μ -only. S is welfare-preferred to M iff $C_S < C_M$.*

The proposition establishes that the welfare-optimal instrument depends on the relative magnitudes of B_V , J_μ , and the fiscal costs, and can differ from the policy that minimizes fiscal cost. Two forces drive this divergence. First, $B_V \geq 0$ reduces the effective cost of the V -instrument below its fiscal cost, making V -only relatively more attractive than a pure fiscal comparison suggests. Second, $J_\mu \geq 0$ raises the effective cost of the μ -instrument above its fiscal cost, making μ -only and the μ -phase of sequencing relatively less attractive. These forces work in the same direction: both favor V over μ , so the welfare ranking is unambiguously tilted toward V -intensive policies relative to the fiscal ranking. When comparing V -only and μ -only interventions (statement (ii)), it merits mentioning that because $B_V(t) \geq 0$ reduces the left side and $J_\mu(t) \geq 0$ increases the right side, this condition can be satisfied even when $c_V > c_\mu$.

An immediate implication of Proposition 7 is if $B_V(t) \geq c_V$ for all $t \in [0, T_V)$, then $\tilde{c}_V^V(t) \leq 0$ for all t and Policy V is strictly welfare-preferred to both M and S . When the per-period welfare bonus from elevated V exceeds the fiscal cost of providing it, every period of the V -phase generates net welfare gains. Cutting the V -phase short to introduce a μ -phase, which carries both a fiscal cost and a welfare penalty, is strictly worse. This condition is satisfied when V -interventions are inexpensive relative to the cooperative surplus they create, as in team-based pay structures where the additional compensation is modest relative to the gains from successful cooperation.

Furthermore, policy M is welfare-improving relative to no intervention only if

$$\frac{R}{\sum_{t=0}^{T_M-1} \delta^t \tilde{c}_\mu^M(t)} > 1,$$

i.e., the discounted value of regime change exceeds the total welfare-adjusted cost of the μ -intervention. No such condition is required for Policy V because $B_V(t) \geq 0$, V -only is welfare-improving for sufficiently small c_V at any $\delta > 0$. Because μ -interventions impose front-loaded welfare costs while their gains are back-loaded, they require the planner to be sufficiently patient. The threshold is increasing in c_μ and in the severity of the μ -penalty. V -interventions face no such asymmetry: $B_V(t) \geq 0$

ensures that welfare is weakly higher in every period of the intervention, so the fiscal cost is the only barrier to welfare improvement.

Lastly, there exist parameter configurations under which $C_S < C_V$ (sequencing has lower fiscal cost than V -only) but $\tilde{C}_V < \tilde{C}_S$. Hence, minimizing fiscal cost can be a misleading criterion. Sequencing has lower fiscal cost than V -only interventions while V -only interventions have lower welfare-adjusted cost than sequencing. When the V -phase periods displaced by sequencing generate welfare bonuses that exceed c_V , the fiscal savings from shortening the V -phase come at a welfare cost. Those periods were net positive for the population. This divergence is sharpest in high-bias environments, where B_V is large (high θ induces substantial effort, so the effort-cost reduction from higher V is valuable) and J_μ is simultaneously large (the effort gap between out-group and in-group matches is wide).

5.4 Distributional Effects

Does the majority or minority benefit more from a successful intervention? At any interior state π with $\mu > -1$, a type-2 (minority) individual's expected per-period welfare is strictly lower than a type-1 (majority) individual's. This is because the average effort of a minority individual is greater than that of a majority individual (Corollary 1). Consequently, the minority gains strictly more from any reduction in aggregate bias, and strictly more from a transition from the high-bias to the low-bias steady state.

The minority interacts out-group more frequently ($\eta_2 < \eta_1$), so a larger fraction of their matches are costly out-group encounters. When aggregate bias falls, the effort-cost reduction is concentrated on out-group matches, and the minority has more of them. This distributional prediction complements Corollary 2. The majority is more biased in steady state, but the minority bears more of the welfare cost. Successful interventions therefore reduces both the level asymmetry (bias gap narrows) and the welfare asymmetry (effort-burden gap narrows).

6 Policy Implications

The model's three main results each carry a distinct policy implication, developed in turn below.

6.1 Effort and the Value of Successful Interaction

Policies that increase V correspond to institutional interventions that increase the returns to successful interaction. This includes team-based pay that ties compensation to cooperative output,

school environments that create shared goals across group lines, or professional settings structured around joint production. A sufficiently large temporary increase in V eliminates the high-bias steady state for the duration of the intervention, driving the economy toward a unique low-bias steady state from any initial condition (Proposition 4). The welfare analysis of Section 5 reinforces this conclusion. V -interventions raise per-period welfare on impact (Proposition 6), and when the welfare bonus exceeds the fiscal cost, V -only is strictly welfare-preferred to any policy involving a μ -phase.

This prediction finds support in studies of integration into high-value settings (Rao, 2019; Boisjoly et al., 2006; Beaman et al., 2009; Hjort, 2014). The model’s mechanism implies a sharper reading of these findings than a directional prediction alone. Proposition 3 implies that the parameter region supporting μ -based tipping shrinks monotonically in V . Environments with higher interaction stakes should therefore exhibit lower steady-state bias but a narrower region in which temporary contact interventions generate lasting change. Consistent with this, Rao (2019) finds that the largest reductions in discriminatory behavior arise specifically in economic exchange tasks, which are precisely the interactions where V is highest and, by the model’s logic, where effort is most strongly induced. A further practical constraint is that if institutional integration is perceived as culturally threatening, individuals may strategically withdraw, endogenously recreating assortative sorting and offsetting the gains from increasing V (Carvalho et al., 2017; Carvalho and Sacks, 2024; Carvalho et al., 2024b).

6.2 Assortative and Disassortative Matching

Policies that raise μ toward disassortative matching correspond to integration mandates, diversity requirements, school assignment policies, and workplace composition rules that increase the frequency of out-group interaction. Conditional on the economy beginning in the basin of attraction of the high-bias steady state, a temporary increase in μ generates a persistent reduction in bias if and only if prevailing average bias lies between the tipping point under the current contact rate and the tipping point under the maximum feasible contact rate.²²

This window-dependence has a direct empirical counterpart. The tipping literature in urban economics documents that neighborhoods and schools exhibit sharp threshold dynamics in racial composition. Once the share of one group crosses a critical level, the social environment transitions

²²Formally, the condition is $\bar{\pi}^0 \in (\bar{\pi}_2^*(\mu^0), \bar{\pi}_2^*(\frac{1-q}{q}))$, where $\frac{1-q}{q}$ is the upper bound on μ imposed by market clearing, $q(1 - \eta_1) = (1 - q)(1 - \eta_2)$.

rapidly and persistently to a segregated state (Schelling, 1971; Card et al., 2008). In the model, these are the kinds of environments where in-group bias feeds back strongly enough on interaction outcomes to support multiple steady states, and where μ -based tipping is feasible.²³

The window condition offers a specific, falsifiable account of cross-setting heterogeneity in contact interventions. Lowe (2021) finds persistent reductions in discrimination following a cooperative cricket program in India, while Mousa (2020) finds that a structurally similar intervention in post-ISIS Iraq produces effects confined to the contact setting. Consistent with the window interpretation, Ghosh et al. (2026) find that a structured youth camp intervention in West Bengal, India generates persistent reductions in in-group bias one year after the camps ended. The model identifies three distinct channels through which such divergence can arise. First, the two populations may have differed in their position relative to the tipping point, with prevailing bias in one setting lying within $(\bar{\pi}_2^*(\mu^0), \bar{\pi}_2^*(\frac{1-q}{q}))$ and in the other lying outside it. Second, the settings may have differed in whether societal feedback is sufficiently sigmoidal to generate tipping dynamics at all, so that the same initial bias level produces a unique stable steady state in one environment and multiple steady states in the other. Third, even when both settings exhibit tipping dynamics, differences in the shape of θ , for instance the size of the relevant social reference group in the community norm enforcement microfoundation, shift the location and width of the tipping window itself, so that identical initial bias levels fall inside the window in one context and outside it in the other. All three channels generate predictions absent from heterogeneity-based explanations: whether a contact program succeeds depends on measurable features of the societal environment, not merely on implementation details.

Steinmayr (2021) finds that communities exposed to refugee resettlement exhibit reduced support for far-right parties when meaningful direct contact occurs, but increased support when exposure occurs without genuine interaction. The model offers a specific account of this asymmetry. Increasing μ raises the rate of out-group contact but does not directly raise the quality of interactions; without a corresponding V , the model predicts that exposure alone is insufficient to cross the tipping threshold, leaving steady-state bias unchanged or inducing reversion. Meaningful contact, by contrast, engages the effort channel and can generate the positive interaction outcomes through which bias is reduced. The model does not account for backlash of the kind documented by Enos (2014), where out-group

²³Interventions that successfully pushed neighborhoods or schools across such thresholds, as in sustained court-ordered desegregation, generated lasting integration (Guryan, 2004); interventions that failed to sustain the mandate were reversed once legal pressure lapsed, with corresponding deterioration in outcomes (Lutz, 2011; Billings et al., 2014).

contact actively increases exclusionary attitudes; that pattern likely reflects mechanisms outside the model’s scope, such as identity threat or status competition. Within interventions that do succeed, however, the model predicts that effects should be asymmetric across majority and minority participants. Ghosh et al. (2026) find, in a randomized youth camp experiment in West Bengal, India, that higher out-group contact intensity backfires among Hindu majority participants but not Muslim minority participants, consistent with the model’s prediction that majority members, facing higher baseline in-group exposure, are more adversely affected by contact interventions that fall short of the tipping threshold.

The welfare analysis provides a further constraint on μ -based interventions. Even when prevailing bias lies within the success window, the welfare J-curve (Proposition 6) implies that μ -interventions impose front-loaded welfare costs on participants before bias-reduction gains materialize. When these costs are severe, as in deeply biased environments where the effort gap between out-group and in-group matches is wide, a planner may prefer V -first sequencing even when μ -only is dynamically feasible (Proposition 7).

The model also implies that contact policy changes the cross-group distribution of bias, not just its average level. Corollary 2 establishes that, under empirically plausible conditions, the majority is strictly more biased than the minority in any interior steady state, reflecting the majority’s higher in-group interaction rate and greater accumulated in-group reinforcement.²⁴ A disassortative matching policy reduces in-group exposure for both groups, but the reduction is proportionally larger for the majority. A uniform increase in μ therefore narrows the bias gap between groups, generating convergence in the distribution of bias alongside a reduction in average bias. This is consistent with evidence that contact effects are often stronger among majority than minority group members (Tropp and Pettigrew, 2005).

6.3 Sequencing

Proposition 5 establishes that a V -first, μ -second policy can shorten the V -phase relative to V -only. The welfare analysis adds a second dimension to this comparison. Proposition 7 shows that the welfare-optimal choice among V -only, μ -only, and sequencing depends on welfare-adjusted costs that incorporate both fiscal expenditure and transition-path welfare effects. The welfare-adjusted ranking can differ from the fiscal ranking. Sequencing may reduce fiscal cost while increasing welfare-

²⁴The conditions are $\mu > -1$, so that out-group contact occurs with positive probability, and $\varepsilon > 0$, so that in-group interactions generate positive reinforcement (ε can be arbitrarily small).

adjusted cost, because the V -phase periods eliminated by sequencing were generating net welfare gains that exceeded their fiscal cost. This divergence is sharpest in high-bias environments, where B_V is large and J_μ is simultaneously large. The welfare analysis therefore implies that sequencing is most attractive when V -interventions are expensive (c_V high), the welfare bonus from elevated V is modest (B_V small), and the J-curve penalty is mild (J_μ small); a configuration more likely in moderately biased environments near the tipping point than in deeply entrenched ones.

7 Concluding Remarks

This paper develops a dynamic model of social interaction to explain why some temporary interventions durably reduce bias while others fail. The key insight is that permanent change requires self-reinforcing societal feedback strong enough to generate a tipping point. When that condition holds, temporarily increasing the rate of out-group contact reduces long-run bias only if prevailing bias lies sufficiently close to the tipping point; otherwise, bias reverts once the intervention ends. Raising the value of successful interaction operates over a broader range of initial conditions, but does so through the same effort response that also weakens the multiplicity needed for temporary contact alone to generate lasting change. Finally, under empirically plausible conditions, the model predicts a majority–minority asymmetry in long-run bias, with majority members more biased than minority members in any interior steady state because differential in-group exposure generates asymmetric reinforcement. A welfare analysis shows that the low-bias steady state strictly welfare-dominates the high-bias steady state, that V -interventions raise per-period welfare on impact while μ -interventions impose a front-loaded welfare penalty, and that the welfare-optimal instrument choice can differ from the one that minimizes fiscal cost.

Together, these results provide a threshold-based framework for understanding the heterogeneous outcomes of contact-based interventions documented in the empirical literature (Paluck et al., 2021; Mousa, 2020; Lowe, 2021, 2025). When initial bias is too entrenched for contact alone to succeed, the framework implies a sequencing logic, where interventions that raise the value of successful interaction can first move the population into the region where a temporary increase in out-group contact can produce lasting change. The formal sequencing result further shows that such a policy sequence can shorten the duration of the value-enhancing intervention relative to a pure- V policy, though not necessarily the total duration of intervention. The welfare analysis refines this prescription. Welfare-adjusted costs, which account for the transition-path welfare effects of each instrument, can

reverse the fiscal case for sequencing.

Having isolated the core mechanism in a tractable setting, the framework suggests four directions for future research. First, the model yields directly testable comparative predictions. In particular, environments with higher stakes for successful interaction should exhibit lower steady-state bias but a smaller region in which temporary contact interventions generate lasting change, while sequenced interventions should outperform contact-only interventions in more deeply entrenched settings. Second, the model takes in-group bias as binary within types. A natural extension is to allow for a distribution of bias levels within each group, which could generate richer distributional dynamics, including whether contact reduces average bias across groups while increasing within-group polarization. Third, endogenizing the contact rate μ is a natural extension. When societal bias is high, individuals may prefer in-group interactions, reducing μ endogenously and reinforcing the aggregate bias that generated the preference for segregation in the first place (Currarini et al., 2009; Gentzkow and Shapiro, 2011; Schelling, 1971). This self-segregation channel would introduce a second feedback loop into the dynamics, potentially amplifying tipping behavior and shrinking the parameter region in which contact-based interventions succeed.²⁵ Fourth, the welfare analysis treats the planner’s objective as utilitarian. The front-loaded welfare costs of μ -interventions may generate political-economy resistance that the utilitarian framework understates. Endogenizing the planner’s instrument choice under political constraints is a natural extension.

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²⁵Carvalho et al. (2025) shows that policies targeting a single identity dimension almost never achieve representative outcomes across all dimensions simultaneously, suggesting that matching mandates aimed at one dimension of identity may generate unintended consequences along others.

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Appendix

Proof of Lemma 1.

Proof. Conditional on i and j being matched with type-bias pairs (τ_i, b_i) and (τ_j, b_j) , i chooses effort

$$e_i^*(b_i, b_j) = \arg \max_{e_i} \left(s(e_i, e_j) + (1 - s(e_i, e_j))[\beta(b_i, b_j) + (1 - \beta(b_i, b_j))(1 - \theta(\bar{\pi}))]V - \frac{1}{2}e_i^2 \right),$$

where $s(e_i, e_j) = e_i + e_j - e_i e_j$. Each j performs an analogous optimization. The corresponding first-order condition, evaluated at $e_i = e_i^*$ and $e_j = e_j^*$, is

$$e_i^* = (1 - e_j^*)(1 - \beta(b_i, b_j))\theta(\bar{\pi})V \iff \frac{e_i^*}{1 - e_j^*} = (1 - \beta(b_i, b_j))\theta(\bar{\pi})V.$$

As $\beta(b_i, b_j) = \beta(b_j, b_i)$, $e_i^* = e_j^*$. Setting $e_i^* = e_j^* = e^*$, substituting, and solving for e^* yields the expression in the lemma. The comparative statics follow immediately. $\square$

The following proposition formalizes the individual-feedback benchmark results.

Proposition A1. *Fix $s(e_i, e_j) = 0$ and $\theta(\bar{\pi}) = 1$ for all e_i, e_j , and $\bar{\pi}$. The biased steady state always exists. If $\varepsilon = 0$ and $\mu > -1$, then the following statements hold.*

- (i) *Almost all initial states converge to the biased steady state iff either $\beta_1 < \frac{1}{2}$ or $\beta_1 = \frac{1}{2}$ and $\beta_0 < 1$.*
- (ii) *If $\beta_0 = 1$, then an unbiased steady state exists, to which almost all initial states converge iff $\beta_1 > \frac{1}{2}$.*
- (iii) *If $\beta_1 > \frac{1}{2}$ and $\beta_0 < 1$, then there exists a unique interior steady state to which almost all states converge. If $\beta_1 = \frac{1}{2}$ and $\beta_0 = 1$, then each initial state converges to $\pi_\tau^\infty = q\pi_1^0 + (1 - q)\pi_2^0$.*

If $\varepsilon > 0$ and $\mu > -1$, then the following statements hold.

- (iv) *If $\frac{\partial F_1(1; \varepsilon)}{\partial \pi_2} \times \frac{\partial F_2(1; \varepsilon)}{\partial \pi_1} \leq 1$, then the biased steady state is the unique steady state.*
- (v) *If $\frac{\partial F_1(1; \varepsilon)}{\partial \pi_2} \times \frac{\partial F_2(1; \varepsilon)}{\partial \pi_1} > 1$, then there exists an interior (asymmetric) steady state, which is componentwise strictly increasing in ε , to which almost all initial conditions converge.*

If $\mu = -1$, then $\pi_\tau^\infty = 1$ when $\varepsilon > 0$ while $\pi_\tau^\infty = \pi_\tau^0$ when $\varepsilon = 0$.

The following two Lemmas are useful for proving Proposition A1

Lemma 5. *Suppose that $s(e_i, e_j) = 0$ for all e_i, e_j , $\theta(\bar{\pi}) = 1$ for all $\bar{\pi}$, $\varepsilon = 0$, and $\mu > -1$. If $(\pi_1^\infty, \pi_2^\infty)$ solves $\Delta\pi_\tau = 0$ for $\tau = 1, 2$, then $\pi_1^\infty = \pi_2^\infty$.*

Proof. Set $s(e_i, e_j) = 0$ for all e_i, e_j and $\theta(\bar{\pi}) = 1$ for all $\bar{\pi}$. Then, by (9),

$$\Delta\pi_\tau = (1 - \eta_\tau) \left[(1 - \pi_\tau) \left((1 - \pi_{-\tau})(1 - \beta_0) + \pi_{-\tau}(1 - \beta_1) \right) - \pi_\tau(1 - \pi_{-\tau})\beta_1 \right], \quad (25)$$

If $\mu > -1$, then $\eta_1, \eta_2 \in (0, 1)$, so $\Delta\pi_\tau = 0$ if and only if

$$\underbrace{(1 - \pi_\tau) \left((1 - \pi_{-\tau})(1 - \beta_0) + \pi_{-\tau}(1 - \beta_1) \right) - \pi_\tau(1 - \pi_{-\tau})\beta_1}_{\equiv F_\tau(\pi_\tau, \pi_{-\tau})} = 0$$

for $\tau = 1, 2$, which is equivalent to $F_1(\pi_1, \pi_2) = 0$ and $F_2(\pi_2, \pi_1) = 0$. Subtracting $F_2(\pi_2, \pi_1)$ from $F_1(\pi_1, \pi_2)$ yields $F_1(\pi_1, \pi_2) - F_2(\pi_2, \pi_1) = \pi_2 - \pi_1$. Thus, $F_1(\pi_1^\infty, \pi_2^\infty) = F_2(\pi_2^\infty, \pi_1^\infty) = 0$ implies $\pi_1^\infty = \pi_2^\infty$. $\square$

Lemma 6. Suppose that $s(e_i, e_j) = 0$ for all e_i, e_j , $\theta(\bar{\pi}) = 1$ for all $\bar{\pi}$, $\varepsilon = 0$, $\mu > -1$, $\beta_0 = 1$, and $\beta_1 = \frac{1}{2}$. Given any initial shares π_1^0 and π_2^0 , there is convergence to $\pi_1^\infty = \pi_2^\infty = q\pi_1^0 + (1-q)\pi_2^0$.

Proof. Given the conditions set forth in the lemma, the law of motion reduces to

$$\begin{pmatrix} \pi_1^{t+1} \\ \pi_2^{t+1} \end{pmatrix} = \underbrace{\begin{pmatrix} 1 - \frac{(1-q)(1+\mu)}{2} & \frac{(1-q)(1+\mu)}{2} \\ \frac{q(1+\mu)}{2} & 1 - \frac{q(1+\mu)}{2} \end{pmatrix}}_{\equiv \mathbf{Q}} \begin{pmatrix} \pi_1^t \\ \pi_2^t \end{pmatrix},$$

with eigenvalues 1 and $\frac{1-\mu}{2} \in [0, 1)$ for $\mu > -1$. A direct calculation confirms that $q\pi_1^{t+1} + (1-q)\pi_2^{t+1} = q\pi_1^t + (1-q)\pi_2^t$, so the weighted average $q\pi_1^t + (1-q)\pi_2^t$ is constant at $q\pi_1^0 + (1-q)\pi_2^0$. As the second eigenvalue lies in $[0, 1)$, the solution converges to a multiple of the unit eigenvector $(1, 1)^T$, so $\pi_1^t - \pi_2^t \rightarrow 0$. Taking $t \rightarrow \infty$ with $\pi_1^\infty = \pi_2^\infty \equiv \pi^\infty$ and using the preserved quantity gives $(q + (1-q))\pi^\infty = q\pi_1^0 + (1-q)\pi_2^0$, hence $\pi_1^\infty = \pi_2^\infty = q\pi_1^0 + (1-q)\pi_2^0$. $\square$

Proof of Proposition A1.

Proof. Set $s(e_i, e_j) = 0$ for all e_i, e_j , $\theta(\bar{\pi}) = 1$ for all $\bar{\pi}$, and $\varepsilon = 0$. First, observe that evaluating (9) at $(\pi_1, \pi_2) = (1, 1)$ yields $\Delta\pi_\tau = 0$ for each τ , so the biased steady state always exists.

To prove statements (i)–(iii), suppose $\varepsilon = 0$ and $\mu > -1$. By Lemma 5, any steady state is symmetric, so $\pi_1^\infty = \pi_2^\infty = \pi^*$ and the steady states are characterized by

$$F(\pi^*) = 0 \iff (1 - \pi^*)[1 - \beta_0 + \pi^*(\beta_0 - 2\beta_1)] = 0. \quad (26)$$

From (26), the bracketed factor of $F(\pi)$ is linear in π , equaling $1 - \beta_0 \geq 0$ at $\pi = 0$ and $1 - 2\beta_1$ at $\pi = 1$. If $\beta_1 < \frac{1}{2}$, the bracket is positive on all of $[0, 1)$, so $F > 0$ and the biased steady state is globally attracting, proving statement (i) (the knife-edge $\beta_0 = 1$, $\beta_1 = \frac{1}{2}$ is Lemma 6). If $\beta_0 = 1$, then $F(\pi) = \pi(1 - \pi)(1 - 2\beta_1)$, so the unbiased steady state exists, and $F < 0$ on $(0, 1)$ whenever $\beta_1 > \frac{1}{2}$, proving statement (ii). If $\beta_0 < 1$ and $\beta_1 > \frac{1}{2}$, the unique interior root $\pi^* = \frac{1 - \beta_0}{2\beta_1 - \beta_0} \in (0, 1)$ is the unique interior steady state, and the sign change of F at π^* together with monotonicity of the joint map implies global convergence from almost all initial conditions, proving statement (iii).

To prove statements (iv) and (v), suppose that $\varepsilon > 0$ and $\mu > -1$. Steady states solve the fixed-point system $\pi_1 = F_1(\pi_2; \varepsilon)$ and $\pi_2 = F_2(\pi_1; \varepsilon)$, where $F_\tau(\cdot; \varepsilon)$ is given by (10). Note that $F_\tau(\cdot; \varepsilon)$ is continuous and strictly increasing on $[0, 1]$ since $\frac{\partial F_\tau(\pi_{-\tau}; \varepsilon)}{\partial \pi_{-\tau}} > 0$ for all $\pi_{-\tau} \in [0, 1]$ by (11). Moreover, the biased steady state $(\pi_1^\infty, \pi_2^\infty) = (1, 1)$ is always a steady state because $F_\tau(1; \varepsilon) = 1$ for each τ .

Define the composite map $H(x; \varepsilon) \equiv F_1(F_2(x; \varepsilon); \varepsilon)$ and $g(x; \varepsilon) \equiv H(x; \varepsilon) - x$. Then $(\pi_1^\infty, \pi_2^\infty)$ is a steady state iff π_1^∞ solves $g(\pi_1^\infty; \varepsilon) = 0$ and $\pi_2^\infty = F_2(\pi_1^\infty; \varepsilon)$. Differentiating H and evaluating at $x = 1$ yields

$$\frac{dH(1; \varepsilon)}{dx} = \frac{\partial F_1(1; \varepsilon)}{\partial \pi_2} \times \frac{\partial F_2(1; \varepsilon)}{\partial \pi_1}.$$

Suppose that $\frac{dH(1; \varepsilon)}{dx} \leq 1$. As $F_\tau(\cdot; \varepsilon)$ is strictly increasing, $H(\cdot; \varepsilon)$ is strictly convex on $[0, 1]$, so $\frac{dH(x; \varepsilon)}{dx} \leq \frac{dH(1; \varepsilon)}{dx} \leq 1$ for all $x \in [0, 1)$. As $\varepsilon > 0$ implies $H(0; \varepsilon) > 0$, we have $g(0; \varepsilon) > 0$, and combined with $g'(x; \varepsilon) < 0$ on $[0, 1)$, it follows that $g(x; \varepsilon) > 0$ on $[0, 1)$, so no interior steady state exists, proving statement (iv).

Now, suppose that $\frac{dH(1;\varepsilon)}{dx} > 1$. Then, $\frac{dg(1;\varepsilon)}{dx} > 0$, so for some small $\delta > 0$, $g(1 - \delta; \varepsilon) = H(1 - \delta; \varepsilon) - (1 - \delta) < 0$. However, $\varepsilon > 0$ implies that $F_\tau(x; \varepsilon) > 0$ for all $x \in [0, 1]$, so $H(0; \varepsilon) > 0$. Hence, $g(0; \varepsilon) > 0$. By continuity, there exists a value $x^* \in (0, 1)$ with $g(x^*; \varepsilon) = 0$, yielding an interior steady state $(\pi_1^\infty, \pi_2^\infty) = (x^*, F_2(x^*; \varepsilon))$. Finally, for any $p \in [0, 1]$, $\frac{\partial F_\tau(p; \varepsilon)}{\partial \varepsilon} > 0$, so both $F_\tau(\cdot; \varepsilon)$ shift up pointwise in ε , and thus so does $H(\cdot; \varepsilon)$. As $H(\cdot; \varepsilon)$ is linear-fractional, $H(x; \varepsilon) - x$ is quadratic with root $x = 1$, hence there is at most one additional (interior) fixed point. Because the interior fixed point is unique, it follows that $x^*(\varepsilon)$ is strictly increasing in ε , and therefore $\pi_2^\infty(\varepsilon) = F_2(x^*(\varepsilon); \varepsilon)$ is strictly increasing as well, proving statement (v).

As $\eta_\tau = 1$ for $\tau = 1, 2$ when $\mu = -1$, the final statement follows immediately from evaluating (9) at $\eta_\tau = 1$. $\square$

The following supporting results for Section 3.3 is useful for proving Proposition 1.

Proposition A2. *Fix $s(e_i, e_j) = 0$ and $\beta(b_i, b_j) = 0$ for all e_i, e_j, b_i , and b_j . The biased steady state exists. Moreover, if $\varepsilon = 0$ and $\mu > -1$, then the following statements hold.*

- (i) *The unbiased steady state exists.*
- (ii) *If $\theta(\bar{\pi})$ is convex, then almost all initial conditions converge to the unbiased steady state.*
- (iii) *If $\theta(\bar{\pi})$ is linear, then initial conditions (π_1^0, π_2^0) converge to the steady state*

$$\pi_1^\infty = \pi_2^\infty = \frac{q^2}{q^2 + (1 - q)^2} \pi_1^0 + \frac{(1 - q)^2}{q^2 + (1 - q)^2} \pi_2^0.$$

If $\varepsilon > 0$ and $\mu > -1$, then the following statements hold.

- (iv) *If $\theta(\bar{\pi})$ is linear, then all initial conditions converge to the biased steady state.*
- (v) *If $\theta(\bar{\pi})$ is convex and $\bar{\gamma} \frac{d\theta(1)}{d\bar{\pi}} > 1$, then in addition to the biased steady state, there is a unique interior steady state $\bar{\pi}^* \in (0, 1)$ to which almost all initial conditions converge. Otherwise, the biased steady state is the unique globally stable steady state.*

If $\mu = -1$, then $\pi_\tau^\infty = 1$ when $\varepsilon > 0$ and $\pi_\tau^\infty = \pi_\tau^0$ when $\varepsilon = 0$.

Proof. Fix $s(e_i, e_j) = 0$ for all e_i, e_j and $\beta(b_i, b_j) = 0$ for all b_i, b_j . Evaluating (15) at $\bar{\pi} = 1$ immediately gives $1 = 1$, so the biased steady state always exists.

To prove statements (i)–(iii), suppose that $\varepsilon = 0$ and $\mu > -1$. At $\varepsilon = 0$, $\gamma_1 = \gamma_2 = 1$, and evaluating at $\bar{\pi} = 0$ gives $0 = 0$, so the unbiased steady state exists, proving statement (i). As $\theta(0) = 0$ and both the unbiased and biased steady states exist, there are no additional steady states. Additionally,

$$\Delta\pi_\tau = (1 - \eta_\tau)(\theta(\bar{\pi}) - \pi_\tau). \tag{27}$$

Both types are pulled each period toward the same target $\theta(\bar{\pi})$. Hence, if $\pi_1 > \pi_2$, the higher type decreases relative to the lower whenever $\theta(\bar{\pi}) \leq \pi_1$, and the lower type increases relative to the higher whenever $\theta(\bar{\pi}) \geq \pi_2$. Thus, the dynamics pull π_1 and π_2 toward the diagonal $\pi_1 = \pi_2$; in particular, any steady state must satisfy $\pi_1 = \pi_2$ and thus $\bar{\pi} = \theta(\bar{\pi})$. For all $\bar{\pi} > 0$, the convexity of $\theta(\bar{\pi})$ coupled with $\theta(0) = 0$ implies that $\theta(\bar{\pi}) - \bar{\pi} < 0$, so there is convergence from almost all initial conditions to the unbiased steady state, , proving statement (ii).²⁶

²⁶If $\theta(\bar{\pi})$ is concave, then by an analogous argument, $\theta(\bar{\pi}) - \bar{\pi} > 0$, so there is convergence from almost all initial conditions to the biased steady state.

Now suppose $\theta(\bar{\pi}) = \bar{\pi}$. The law of motion (13) reduces to

$$\begin{pmatrix} \pi_1^{t+1} \\ \pi_2^{t+1} \end{pmatrix} = \underbrace{\begin{pmatrix} 1 - (1 - \eta_1)(1 - q) & (1 - \eta_1)(1 - q) \\ (1 - \eta_2)q & 1 - (1 - \eta_2)q \end{pmatrix}}_{\equiv \mathbf{Q}} \begin{pmatrix} \pi_1^t \\ \pi_2^t \end{pmatrix},$$

with eigenvalues 1 and $(1 - q)\eta_1 + q\eta_2 \in (0, 1)$ for $\mu > -1$. A direct calculation confirms that $q(1 - \eta_2)\pi_1^{t+1} + (1 - q)(1 - \eta_1)\pi_2^{t+1} = q(1 - \eta_2)\pi_1^t + (1 - q)(1 - \eta_1)\pi_2^t$, so the linear combination on the left is constant at its initial value, and since the second eigenvalue lies in $(0, 1)$, the solution converges to a multiple of the unit eigenvector $(1, 1)^T$ so $\pi_1^t - \pi_2^t \rightarrow 0$, proving statement (iii).

To prove statements (iv) and (v), suppose that $\varepsilon > 0$ and $\mu > -1$. In this case $(\gamma_1, \gamma_2) \in (0, 1)^2$. Now suppose that $\theta(\bar{\pi})$ is linear. At $\pi_1^0 = \pi_2^0 = 0$ (so $\bar{\pi} = 0$),

$$[q\gamma_1 + (1 - q)\gamma_2](0 - 1) + 1 > 0.$$

As the biased steady state exists, it immediately follows (by an analogous argument to statement (ii)) that all initial conditions converge to the biased steady state, proving statement (iv).²⁷

Now suppose that $\theta(\bar{\pi})$ is convex and $[q\gamma_1 + (1 - q)\gamma_2] \frac{d\theta(1)}{d\bar{\pi}} > 1$. Define $H(\bar{\pi}) \equiv [q\gamma_1 + (1 - q)\gamma_2](\theta(\bar{\pi}) - 1) + 1$ and $g(\bar{\pi}) \equiv H(\bar{\pi}) - \bar{\pi}$. As $\varepsilon > 0$ and $(\gamma_1, \gamma_2) \in (0, 1)^2$, $g(0) > 0$ and $g(1) = 0$. The assumed condition gives $\frac{dg(1)}{d\bar{\pi}} > 0$, so $g < 0$ in a left-neighborhood of 1; by the intermediate value theorem there exists $\bar{\pi}^* \in (0, 1)$ with $g(\bar{\pi}^*) = 0$. The convexity of θ implies convexity of g , so $\bar{\pi}^*$ is unique and $\frac{dg(\bar{\pi}^*)}{d\bar{\pi}} < 0$, establishing global stability. Uniqueness of the constituent π_1^*, π_2^* follows because the right-hand side of (14) depends on π_τ only through $\bar{\pi}^*$, proving statement (v). If the slope condition fails, $g(\bar{\pi}) > 0$ for all $\bar{\pi} < 1$ and the biased steady state is the unique globally stable steady state.

If $\mu = -1$, then as is the case in Proposition A1, evaluating (13) at $\mu = -1$ (so $\eta_\tau = 1$ for $\tau = 1, 2$) yields the last statement. $\square$

Proof of Proposition 1.

Proof. Fix $s(e_i, e_j) = 0$ for all e_i, e_j and $\beta(b_i, b_j) = 0$ for all b_i, b_j . By the exact same argument as Proposition A2, the biased steady state exists. Suppose that $\varepsilon > 0$ and $\mu > -1$. Define

$$H(\bar{\pi}) = [q\gamma_1 + (1 - q)\gamma_2](\theta(\bar{\pi}) - 1) + 1$$

as the right-hand side of (15) and $g(\bar{\pi}) = H(\bar{\pi}) - \bar{\pi}$. Then, $H(0) = 1 - [q\gamma_1 + (1 - q)\gamma_2] > 0$ (as $(\gamma_1, \gamma_2) \in (0, 1)^2$), so the unbiased steady state no longer exists. As $\frac{dH(\bar{\pi})}{d\bar{\pi}} = [q\gamma_1 + (1 - q)\gamma_2] \frac{d\theta(\bar{\pi})}{d\bar{\pi}}$, additional interior steady states exist iff $g(\bar{\pi}) < 0$ for some $\bar{\pi} \in (0, 1)$. Because $g(0) = 1 - [q\gamma_1 + (1 - q)\gamma_2] > 0$ and $g(1) = 0$, this condition requires that $\frac{dg(\bar{\pi})}{d\bar{\pi}}$ becomes positive somewhere, i.e., $\frac{d\theta(\bar{\pi})}{d\bar{\pi}} > [q\gamma_1 + (1 - q)\gamma_2]^{-1}$ for some $\bar{\pi}$. Interior steady states therefore exist iff $\frac{d\theta(\bar{x})}{d\bar{\pi}} > [q\gamma_1 + (1 - q)\gamma_2]^{-1}$, where $\bar{x}$ maximizes $\frac{d\theta(\bar{\pi})}{d\bar{\pi}}$ on $(0, 1)$. As θ is sigmoidal, g has exactly one local minimum followed by one local maximum on $(0, 1)$; together with $g(0) > 0$, $g(1) = 0$, and $\frac{dg(1)}{d\bar{\pi}} < 0$, this permits exactly two interior zeros when the condition holds. If this condition holds, there are three steady states: a stable interior steady state $\bar{\pi}^*$, an unstable interior tipping point $\bar{\pi}_U^* > \bar{\pi}^*$, and the biased steady state, with $g(\bar{\pi}) > 0$ on $(0, \bar{\pi}^*)$, $g(\bar{\pi}) < 0$ on $(\bar{\pi}^*, \bar{\pi}_U^*)$, and $g(\bar{\pi}) > 0$ on $(\bar{\pi}_U^*, 1)$, giving the respective basins of attraction. If this condition fails, $g(\bar{\pi}) > 0$ for all $\bar{\pi} \in [0, 1)$ and the biased steady state is the unique steady state, proving statement (i).

²⁷Allowing weak concavity does not affect this argument.

Suppose that the conditions of statement (i) hold. Consider $\mu \in (-1, \frac{1-q}{q}]$

Note that

$$\gamma_\tau = \frac{1 - \eta_\tau}{1 - \eta_\tau(1 - \varepsilon)}$$

is decreasing in η_τ and thus increasing in μ . Therefore, if $H(\bar{\pi}_U^*(\mu)) = \bar{\pi}_U^*(\mu)$, then for all $\mu' > \mu$, evaluating γ_1 and γ_2 at μ' and $\bar{\pi} = \bar{\pi}_U^*(\mu)$ yields

$$[q\gamma(\eta_1(\mu'), \varepsilon) + (1 - q)\gamma(\eta_2(\mu'), \varepsilon)](\theta(\bar{\pi}_U^*(\mu)) - 1) + 1 < \bar{\pi}_U^*(\mu),$$

so $\bar{\pi}_U^*(\mu') > \bar{\pi}_U^*(\mu)$.

I now verify that $\bar{\pi}^*(\mu)$ is strictly decreasing in μ . At a stable interior steady state, $g(\bar{\pi}^*(\mu)) = 0$ and $\frac{\partial g(\bar{\pi})}{\partial \bar{\pi}} < 0$. Differentiating implicitly yields

$$\frac{d\bar{\pi}^*}{d\mu} = -\frac{\partial g / \partial \mu}{\partial g / \partial \bar{\pi}^*}.$$

Because $g(\bar{\pi}) = [q\gamma_1 + (1 - q)\gamma_2](\theta(\bar{\pi}) - 1) + 1 - \bar{\pi}$ and $\theta(\bar{\pi}) - 1 < 0$ for $\bar{\pi} \in (0, 1)$, an increase in μ lowers g pointwise, so $\frac{\partial g(\bar{\pi})}{\partial \mu} < 0$. Combined with $\frac{\partial g(\bar{\pi}^*)}{\partial \bar{\pi}} < 0$ at a stable fixed point, $\frac{d\bar{\pi}^*}{d\mu} < 0$. Hence, $\bar{\pi}^*(\mu)$ is strictly decreasing in μ .

Now, consider an initial $\mu = \mu^0$ and $\bar{\pi}^0 > \bar{\pi}_U^*(\mu^0)$. In this case, under $\mu = \mu^0$, the trajectory remains in the basin of attraction of the biased steady state and converges toward it. As $\bar{\pi}_U^*(\mu)$ is continuous in μ (by the implicit function theorem, since $\frac{dH(\bar{\pi}_U^*(\mu))}{d\bar{\pi}} > 1$ implies $\frac{\partial g(\bar{\pi}_U^*)}{\partial \bar{\pi}} \neq 0$) and strictly increasing in μ , and $\bar{\pi}^0 < \bar{\pi}_U^*(\frac{1-q}{q})$ by hypothesis, there exists $\mu' \in (\mu^0, \frac{1-q}{q})$ such that $\bar{\pi}^0 < \bar{\pi}_U^*(\mu')$, so at $\mu = \mu'$, $\bar{\pi}^0$ is in the basin of attraction for the lower steady state.

Under $\mu = \mu'$, the trajectory converges to the lower interior steady state, so there exists $t' > 0$ such that $\bar{\pi}^{t'} < \bar{\pi}_U^*(\mu^0)$ (as $\bar{\pi}^*(\mu') < \bar{\pi}^*(\mu^0) < \bar{\pi}_U^*(\mu^0)$, where the first inequality follows from $\frac{d\bar{\pi}^*}{d\mu} < 0$ and $\mu' > \mu^0$, and the second from above. Therefore, returning μ to μ^0 implies that there will be persistent convergence of $\bar{\pi}^t$ to $\bar{\pi}^*(\mu^0)$, proving statement (ii). $\square$

Proof of Lemma 2.

Proof. Recall from Section 2.2 that

$$\rho_{b_i, b_j} = s^*(b_i, b_j) + (1 - s^*(b_i, b_j))[\beta(b_i, b_j) + (1 - \beta(b_i, b_j))(1 - \theta(\bar{\pi}))].$$

Because efforts are symmetric in the equilibrium of the social interaction stage game, letting $e_i^*(b_i, b_j) = e_j^*(b_i, b_j) = e^*$ (suppressing the bias values),

$$s^*(b_i, b_j) = e^* + e^* - e^* \times e^* = 1 - (1 - e^*)^2.$$

Now, consider the probability of a negative interaction between i and j :

$$\begin{aligned} 1 - \rho_{b_i, b_j} &= 1 - s^*(b_i, b_j) - (1 - s^*(b_i, b_j))[\beta(b_i, b_j) + (1 - \beta(b_i, b_j))(1 - \theta(\bar{\pi}))] \\ &= (1 - e^*)^2(1 - \beta(b_i, b_j))\theta(\bar{\pi}), \end{aligned}$$

so

$$\rho_{b_i, b_j} = 1 - (1 - e^*)^2 (1 - \beta(b_i, b_j)) \theta(\bar{\pi}),$$

Substituting (5) for e^* yields

$$\rho_{b_i, b_j} = 1 - \frac{(1 - \beta(b_i, b_j)) \theta(\bar{\pi})}{[1 + (1 - \beta(b_i, b_j)) \theta(\bar{\pi}) V]^2}.$$

Differentiating this expression with respect to $\bar{\pi}$ yields

$$\frac{\partial \rho_{b_i, b_j}}{\partial \bar{\pi}} = -\frac{d\theta(\bar{\pi})}{d\bar{\pi}} \times \frac{(1 - \beta(b_i, b_j)) [1 - (1 - \beta(b_i, b_j)) \theta(\bar{\pi}) V]}{[1 + (1 - \beta(b_i, b_j)) \theta(\bar{\pi}) V]^3}.$$

As $\frac{d\theta(\bar{\pi})}{d\bar{\pi}} > 0$ and $(1 - \beta(b_i, b_j)) \in [0, 1]$, $\frac{\partial \rho_{b_i, b_j}}{\partial \bar{\pi}}$ is monotonic in $\bar{\pi}$ if and only if

$$1 - (1 - \beta(b_i, b_j)) \theta(\bar{\pi}) V$$

is monotonic; i.e., if at $\bar{\pi} = 1$,

$$\begin{aligned} 1 - (1 - \beta(b_i, b_j)) V &\geq 0 \\ \iff V &\leq \frac{1}{1 - \beta(b_i, b_j)} \equiv \hat{V}, \end{aligned}$$

proving the Lemma. □

Proof of Lemma 3.

Proof. Suppose that $\mu > -1$. Recall by (6) that

$$\Delta \pi_\tau = (1 - \eta_\tau) [1 - \bar{\rho}_\tau(\pi) - \pi_\tau] + \eta_\tau (1 - \pi_\tau) \varepsilon,$$

which when evaluated at the biased steady state $(\pi_1, \pi_2) = (1, 1)$ yields

$$\Delta \pi_\tau = -(1 - \eta_\tau) \bar{\rho}_\tau(1).$$

Hence, for $\mu > -1$ (so $\eta_\tau < 1$), the biased steady state exists if and only if $\bar{\rho}_\tau(1) = 0$, where by the proof of Lemma 2,

$$\bar{\rho}_\tau(1) = \rho_{1,1} = 1 - \frac{\theta(1)}{[1 + \theta(1)V]^2} = 1 - \frac{1}{(1 + V)^2} = \frac{V(2 + V)}{(1 + V)^2} > 0.$$

Therefore, the biased steady state does not exist.

Now, evaluating (6) at the unbiased steady state $(\pi_1, \pi_2) = (0, 0)$ yields

$$\Delta \pi_\tau = -(1 - \eta_\tau)(1 - \bar{\rho}_\tau(0)) - \eta_\tau \varepsilon.$$

Hence, for $\mu > -1$, the unbiased steady state exists if and only if $\bar{\rho}_\tau(0) = 1$ and $\varepsilon = 0$, where by the proof of Lemma 2,

$$\bar{\rho}_\tau(0) = \rho_{0,0} = 1 - \frac{(1 - \beta_0) \theta(0)}{[1 + (1 - \beta_0) \theta(0)V]^2} = 1$$

Thus, the unbiased steady state exists whenever $\varepsilon = 0$ and $\mu > -1$. □

Proof of Proposition 2.

Proof. Fix $\mu > -1$. Recall by (6) that

$$\Delta\pi_\tau = (1 - \eta_\tau)[1 - \bar{\rho}_\tau(\pi) - \pi_\tau] + \eta_\tau(1 - \pi_\tau)\varepsilon$$

where $\bar{\rho}_\tau(\pi)$ is the (type- τ) expected probability of a positive interaction under equilibrium effort.

Fix a realized pair of biases (b_i, b_j) and write $\beta \equiv \beta(b_i, b_j) \in [0, 1]$ and $\theta \equiv \theta(\bar{\pi})$. From Lemma 1, equilibrium effort satisfies

$$1 - s^*(b_i, b_j) = (1 - e^*(b_i, b_j))^2 = \frac{1}{(1 + (1 - \beta)\theta V)^2},$$

so the match-level negative-interaction probability is

$$m_\beta(\bar{\pi}) \equiv \Pr(\text{negative} \mid b_i, b_j) = \frac{(1 - \beta)\theta(\bar{\pi})}{(1 + (1 - \beta)\theta(\bar{\pi})V)^2}. \quad (28)$$

Equivalently, $\rho_{b_i, b_j}(\bar{\pi}) = 1 - m_\beta(\bar{\pi})$.

Define

$$\tilde{p}'(\bar{\pi}) := \frac{d\theta(\bar{\pi})}{d\bar{\pi}} \times \max_{\beta \in [0, 1]} \frac{(1 - \beta)|1 - (1 - \beta)\theta(\bar{\pi})V|}{(1 + (1 - \beta)\theta(\bar{\pi})V)^3}.$$

Define $z = (1 - \beta)\theta(\bar{\pi})V$, so the above can be represented by the equivalent optimization problem

$$\frac{1}{\theta(\bar{\pi})V} \times \max_{z \in [0, \theta(\bar{\pi})V]} \frac{z|1 - z|}{(1 + z)^3}.$$

Solving this maximization problem yields

$$(1 - \beta)\theta V = z^* = \begin{cases} 2 - \sqrt{3} & \text{if } \theta(\bar{\pi})V \leq 2 + \sqrt{3} \\ 2 + \sqrt{3} & \text{if } \theta(\bar{\pi})V > 2 + \sqrt{3}. \end{cases}$$

As $\frac{z^*|1 - z^*|}{(1 + z^*)^3} = \frac{\sqrt{3}}{18}$ for both $z \leq 1$ and $z > 1$, $|\frac{dm_\beta}{d\bar{\pi}}| \leq \tilde{p}'(\bar{\pi})$. Define $\tilde{P}' := \sup_{\bar{\pi} \in [0, 1]} \tilde{p}'(\bar{\pi})$.

Now, denote by

$$\bar{\Delta} = \sup_{\bar{\pi} \in [0, 1]} \max_{\beta, \beta' \in \{0, \beta_1, \beta_0\}} |m_\beta(\bar{\pi}) - m_{\beta'}(\bar{\pi})| \geq 0$$

the largest gap between the $m_\beta(\bar{\pi})$ terms across all bias configurations. Expanding $M_\tau(\pi) = 1 - \bar{\rho}_\tau(\pi)$ yields

$$M_\tau(\pi) = (1 - \pi_\tau)(1 - \pi_{-\tau})m_{\beta_0} + [(1 - \pi_\tau)\pi_{-\tau} + \pi_\tau(1 - \pi_{-\tau})]m_{\beta_1} + \pi_\tau\pi_{-\tau}m_{\beta_2}.$$

Differentiating this expression and letting $q_\tau = \lambda(I_\tau)$ yields

$$\frac{\partial M_\tau}{\partial \pi_\tau} = \underbrace{(1 - \pi_{-\tau})(m_{\beta_1} - m_{\beta_0}) + \pi_{-\tau}(m_{\beta_2} - m_{\beta_1})}_{W_\tau^{\text{own}}} + q_\tau \frac{\partial M_\tau}{\partial \bar{\pi}}$$

$$\frac{\partial M_\tau}{\partial \pi_{-\tau}} = \underbrace{(1 - \pi_\tau)(m_{\beta_1} - m_{\beta_0}) + \pi_\tau(m_{\beta_2} - m_{\beta_1})}_{W_\tau^{\text{cross}}} + q_{-\tau} \frac{\partial M_\tau}{\partial \bar{\pi}}.$$

Each W_τ^{own} is a convex combination of $(m_{\beta_1} - m_{\beta_0})$ and $(m_{\beta_2} - m_{\beta_1})$, so $|W_\tau^{\text{own}}| \leq \bar{\Delta}$. Likewise, $|W_\tau^{\text{cross}}| \leq \bar{\Delta}$. By the triangle inequality and the fact that $|\frac{dm_\beta}{d\bar{\pi}}| \leq \tilde{p}'(\bar{\pi})$,

$$\sum_{k=1}^2 \left| \frac{\partial M_\tau}{\partial \pi_k} \right| \leq |W_\tau^{\text{own}}| + |W_\tau^{\text{cross}}| + \sum_{k=1}^2 q_k \left| \frac{\partial M_\tau}{\partial \bar{\pi}} \right| \leq 2\bar{\Delta} + \tilde{P}'. \quad (29)$$

At a steady state, (6) implies

$$0 = (1 - \eta_\tau)[M_\tau(\pi) - \pi_\tau] + \eta_\tau(1 - \pi_\tau)\varepsilon,$$

which can be rearranged to

$$\pi_\tau = \frac{(1 - \eta_\tau)M_\tau(\pi) + \eta_\tau\varepsilon}{1 - \eta_\tau(1 - \varepsilon)} \equiv T_\tau(\pi), \quad (30)$$

$\tau \in \{1, 2\}$. Thus, steady states are fixed points of the mapping $T = (T_1, T_2) : [0, 1]^2 \rightarrow [0, 1]^2$. Because $T_\tau(\pi) = \gamma_\tau M_\tau(\pi) + (1 - \gamma_\tau)\varepsilon$,

$$\sum_{k=1}^2 \left| \frac{\partial T_\tau}{\partial \pi_k} \right| = \gamma(\eta_\tau, \varepsilon) \sum_{k=1}^2 \left| \frac{\partial M_\tau}{\partial \pi_k} \right| \leq \gamma(\eta_\tau, \varepsilon)(2\bar{\Delta} + \tilde{P}'). \quad (31)$$

As $\gamma(\eta_\tau, \varepsilon)$ is strictly decreasing in η and $\eta_1 \geq \eta_2$, the maximum row sum is attained at $\tau = 2$. Now, assume that $2\bar{\Delta} + \tilde{P}' < \gamma(\eta_2, \varepsilon)^{-1}$. Then (31) implies that

$$\max_{\tau \in \{1, 2\}} \sum_{k=1}^2 \left| \frac{\partial T_\tau}{\partial \pi_k} \right| < 1$$

uniformly on $[0, 1]^2$, so T is a global contraction. By Banach's fixed-point theorem, T has a unique fixed point in $[0, 1]^2$. Lemma 3 rules out $(1, 1)$ for $\mu > -1$, so the unique steady state lies in $[0, 1)^2$, proving statement (i).

Lastly, assume $\varepsilon > 0$ and that $\frac{dm_{\beta_0}(\bar{\pi})}{d\bar{\pi}}$ is single-peaked on $(0, 1)$, attaining its unique maximum at $\bar{\pi}_M \in (0, 1)$. Let $M_\tau(\pi) = m_{\beta_0} + R_\tau(\pi)$, where $|R_\tau(\pi)| \leq \bar{\Delta}$ for all $\pi \in [0, 1]^2$ and $\bar{\gamma} = q\gamma_1 + (1 - q)\gamma_2$. Define the aggregate map

$$\Psi(\bar{\pi}) = \bar{\gamma}m_{\beta_0}(\bar{\pi}) + \sum_{\tau} q_\tau \frac{\eta_\tau \varepsilon}{1 - \eta_\tau(1 - \varepsilon)} > 0,$$

so that at any steady state, $\bar{\pi} = \Psi(\bar{\pi}) + \mathcal{E}(\pi)$ where $\mathcal{E}(\pi)$ is an error term such that $|\mathcal{E}(\pi)| \leq \bar{\Delta}$ uniformly. As $\frac{d\Psi}{d\bar{\pi}} = \bar{\gamma} \times \frac{dm_{\beta_0}}{d\bar{\pi}}$ and $\frac{dm_{\beta_0}}{d\bar{\pi}}$ is single-peaked at $\bar{\pi}_M$ by hypothesis, $\frac{d\Psi}{d\bar{\pi}}$ is also single-peaked at $\bar{\pi}_M$. The conditions

$$\frac{dm_{\beta_0}(\bar{\pi}_M)}{d\bar{\pi}} > \bar{\gamma}^{-1}, \quad \tilde{p}'(0), \tilde{p}'(1) < \bar{\gamma}^{-1}$$

imply that $\frac{d\Psi}{d\bar{\pi}} < 1$ near $\bar{\pi} = 0$ and $\bar{\pi} = 1$ (using $\frac{dm_{\beta_0}}{d\bar{\pi}} \leq \tilde{p}'$ at the endpoints), while $\frac{d\Psi}{d\bar{\pi}} > 1$ at $\bar{\pi}_M$. As

$$\Psi(0) = \sum_{\tau} q_\tau \frac{\eta_\tau \varepsilon}{1 - \eta_\tau(1 - \varepsilon)}$$

Ψ lies above the diagonal near 0 and $\Psi(1) < 1$ (as $m_{\beta_0}(1) < 1$ and $\bar{\gamma} < 1$), the graph of Ψ crosses the 45-degree line exactly three times at $\bar{\pi}_1^* < \bar{\pi}_2^* < \bar{\pi}_3^*$.

For each $\bar{\pi}_k^*$, the fixed-point equations $\pi_\tau = T_\tau(\pi)$ subject to $q\pi_1 + (1-q)\pi_2 = \bar{\pi}_k^*$ reduce to a single equation in one free variable. By the monotonicity and continuity of T_τ , this equation has at least one solution, yielding a steady state $\pi^{(k)} \in (0, 1)^2$.

More precisely, at each crossing $\bar{\pi}_k^*$, define the two-dimensional map $\Phi(\pi) = T(\pi) - \pi$. At $\bar{\Delta} = 0$ (i.e., $\beta_1 = \beta_0$ and $\beta_2 = \beta_0$), the system reduces exactly to the one-dimensional map. $M_\tau(\pi)$ depends on π only through $\bar{\pi}$, so the fixed-point equations collapse to $\bar{\pi} = \Psi(\bar{\pi})$ together with $\pi_\tau = \gamma_\tau m_{\beta_0}(\bar{\pi}) + (1 - \gamma_\tau)$. Each of the three aggregate crossings then yields a unique $(\pi_1^{(k)}, \pi_2^{(k)})$ at which $\Phi = 0$. At the two stable crossings ($k = 1, 3$), $\frac{d\Psi}{d\bar{\pi}} < 1$, and at the unstable crossing ($k = 2$), $\frac{d\Psi}{d\bar{\pi}} > 1$. In each case $D\Phi$ is nonsingular at the corresponding fixed point. By the implicit function theorem, each fixed point persists under small perturbations of the bias parameters. As $\bar{\Delta}$ is continuous in $(\beta_0, \beta_1, \beta_2)$ and equals zero when $\beta_0 = \beta_1 = \beta_2$, the three fixed points persist for all $\bar{\Delta} < \hat{\Delta}$ for some $\hat{\Delta} > 0$.²⁸

Let $g(\bar{\pi}) = \Psi(\bar{\pi}) - \bar{\pi}$. By the above analysis, g has exactly three zeros at $\bar{\pi}_1^* < \bar{\pi}_2^* < \bar{\pi}_3^*$. On the interval $[\bar{\pi}_1^*, \bar{\pi}_2^*]$, g is strictly negative (as Ψ crosses the diagonal from above at $\bar{\pi}_1^*$ and from below at $\bar{\pi}_2^*$). The error $\mathcal{E}(\pi)$ perturbs each crossing level by at most $\bar{\Delta}$. Define

$$d_1 = - \min_{\bar{\pi} \in [\bar{\pi}_1^*, \bar{\pi}_2^*]} g(\bar{\pi}) > 0.$$

Similarly, on $[\bar{\pi}_2^*, \bar{\pi}_3^*]$, g is strictly positive. Define

$$d_2 = \max_{\bar{\pi} \in [\bar{\pi}_2^*, \bar{\pi}_3^*]} g(\bar{\pi}) > 0.$$

The full steady-state system requires $g(\bar{\pi}) = \mathcal{E}(\pi)$, where $|\mathcal{E}(\pi)| \leq \bar{\gamma}\bar{\Delta}$ uniformly.

Assume $\bar{\Delta} < \min(d_1/\bar{\gamma}, d_2/\bar{\gamma}, (1-\bar{\gamma})/\bar{\gamma}, 1 - m_{\beta_0}(1)) \equiv \hat{\Delta}$. At $\bar{\pi} = 0$, $g(0) = 1 - \bar{\gamma} > 0$. At the minimizer $\bar{\pi}_{\min} \in (\bar{\pi}_1^*, \bar{\pi}_2^*)$, $g(\bar{\pi}_{\min}) = -d_1 < -\bar{\gamma}\bar{\Delta}$. Hence, for any $\mathcal{E}$ with $|\mathcal{E}| \leq \bar{\gamma}\bar{\Delta}$, the $g(\bar{\pi}) - \mathcal{E}$ satisfies $g(0) - \mathcal{E} \geq (1 - \bar{\gamma}) - \bar{\gamma}\bar{\Delta} > 0$ and $g(\bar{\pi}_{\min}) - \mathcal{E} \leq -d_1 + \bar{\gamma}\bar{\Delta} < 0$. By the intermediate value theorem, there exists a root $\hat{\pi}_1 \in (0, \bar{\pi}_{\min})$. At the maximizer $\bar{\pi}_{\max} \in (\bar{\pi}_2^*, \bar{\pi}_3^*)$, $g(\bar{\pi}_{\max}) = d_2 > \bar{\gamma}\bar{\Delta}$. As $g(\bar{\pi}_{\min}) - \mathcal{E} < 0$ and $g(\bar{\pi}_{\max}) - \mathcal{E} > d_2 - \bar{\gamma}\bar{\Delta} > 0$, $g(\bar{\pi}) - \mathcal{E}$ changes sign on $[\bar{\pi}_{\min}, \bar{\pi}_{\max}]$, yielding a root $\hat{\pi}_2$. Finally, $g(1) = \Psi(1) - 1 < 0$ (as $m_{\beta_0}(1) < 1$ and $\bar{\gamma} < 1$), so $g(1) - \mathcal{E} \leq g(1) + \bar{\gamma}\bar{\Delta}$. Because $g(1) = -\bar{\gamma}(1 - m_{\beta_0}(1)) < 0$, this remains strictly negative for $\bar{\Delta} < 1 - m_{\beta_0}(1)$. As $g(\bar{\pi}_{\max}) - \mathcal{E} > 0$, $g(\bar{\pi}) - \mathcal{E}$ changes sign on $[\bar{\pi}_{\max}, 1]$, yielding a third root $\hat{\pi}_3$.

For each aggregate crossing $\hat{\pi}_k$, the fixed-point equations $\pi_\tau = T_\tau(\pi)$ subject to $q\pi_1 + (1-q)\pi_2 = \hat{\pi}_k$ reduce to a single equation in one free variable. By the monotonicity and continuity of T_τ , and because T_τ maps $[0, 1]^2$ into $(0, 1)^2$ (since $\varepsilon > 0$ and $V > 0$), this equation has at least one solution, yielding a steady state $\pi^{(k)} \in (0, 1)^2$. The three crossings $\hat{\pi}_1 < \hat{\pi}_2 < \hat{\pi}_3$ are distinct, so the corresponding steady states are also distinct, proving statement (ii). $\square$

Proposition A3. *Suppose $\varepsilon > 0$ and parameters are contained in Ω for all $\mu \in [\mu^0, \frac{1-q}{q}]$. If $\bar{\pi}^0 \in (\bar{\pi}_2^*(\mu^0), \bar{\pi}_2^*(\frac{1-q}{q}))$, then there exists a $\mu' > \mu^0$ such that temporarily increasing μ^0 to μ' yields a persistent reduction in long-run bias with convergence to $\bar{\pi}_1^*(\mu^0)$.*

²⁸The binding constraint is not proximity of β_1 to β_0 per se, but rather that the amplitude of the sigmoidal nonlinearity in Ψ exceeds the perturbation introduced by heterogeneous β values. Concretely, $\hat{\Delta}$ is bounded below by $\min_k |1 - \frac{d\Psi(\bar{\pi}_k^*)}{d\bar{\pi}}|$ scaled by a Lipschitz constant. When the three crossings of Ψ with the diagonal are well separated and steep, the fixed points survive large differences between β_0 and β_1 . The condition $\bar{\Delta} < \hat{\Delta}$ is therefore least restrictive precisely when tipping dynamics are most pronounced.

Proof. From the proof of Proposition 2, the aggregate map is

$$\Psi(\bar{\pi}; \mu) = \bar{\gamma}(\mu)m_{\beta_0}(\bar{\pi}) + \sum_{\tau} q_{\tau} \frac{\eta_{\tau}(\mu)\varepsilon}{1 - \eta_{\tau}(\mu)(1 - \varepsilon)}.$$

As $1 - \gamma_{\tau} = \frac{\eta_{\tau}\varepsilon}{(1 - \eta_{\tau}(1 - \varepsilon))}$, summing gives $\sum_{\tau} q_{\tau}(1 - \gamma_{\tau}) = 1 - \bar{\gamma}$, so

$$\Psi(\bar{\pi}; \mu) = 1 - \bar{\gamma}(\mu)(1 - m_{\beta_0}(\bar{\pi})).$$

Define $g(\bar{\pi}; \mu) \equiv \Psi(\bar{\pi}; \mu) - \bar{\pi}$. Because $\bar{\gamma}(\mu)$ is strictly increasing in μ (since η_{τ} is strictly decreasing in μ and $\gamma_{\tau} = \frac{(1 - \eta_{\tau})}{(1 - \eta_{\tau}(1 - \varepsilon))}$ is strictly decreasing in η_{τ}), and $m_{\beta_0}(\bar{\pi}) < 1$ for all $\bar{\pi} \in [0, 1]$ (as $V > 0$), it follows that

$$\frac{\partial g(\bar{\pi}; \mu)}{\partial \mu} = -\frac{d\bar{\gamma}}{d\mu}(1 - m_{\beta_0}(\bar{\pi})) < 0$$

for all $\bar{\pi} \in [0, 1)$.

At the unstable crossing $\bar{\pi}_2^*$, the map Ψ crosses the diagonal from below, so $\frac{\partial g(\bar{\pi}; \mu)}{\partial \bar{\pi}} > 0$. The implicit function theorem applied to $g(\bar{\pi}_2^*(\mu); \mu) = 0$ yields

$$\frac{d\bar{\pi}_2^*}{d\mu} = -\frac{\partial g/\partial \mu}{\partial g/\partial \bar{\pi}} > 0,$$

so $\bar{\pi}_2^*(\mu)$ is strictly increasing in μ . At $\bar{\pi}_1^*$, the map Ψ crosses from above, so $\partial g/\partial \bar{\pi} < 0$. By the same argument, $\frac{d\bar{\pi}_1^*}{d\mu} < 0$, so $\bar{\pi}_1^*(\mu)$ is strictly decreasing in μ .

The remainder of the argument follows the proof of Proposition 1. As $\bar{\pi}_2^*(\mu)$ is continuous and strictly increasing in μ with $\bar{\pi}^0 < \bar{\pi}_2^*(\frac{1-q}{q})$ by hypothesis, there exists $\mu' \in (\mu^0, \frac{1-q}{q})$ such that $\bar{\pi}^0 < \bar{\pi}_2^*(\mu')$. Thus, $\bar{\pi}_1^*(\mu') < \bar{\pi}_1^*(\mu^0) < \bar{\pi}_2^*(\mu^0) < \bar{\pi}^0 < \bar{\pi}_2^*(\mu')$, and the trajectory from $\bar{\pi}^0$ under μ' lies in the basin of attraction of $\bar{\pi}_1^*(\mu')$ and converges toward it. Because $\bar{\pi}_1^*(\mu') < \bar{\pi}_1^*(\mu^0) < \bar{\pi}_2^*(\mu^0)$ (using strict decrease of $\bar{\pi}_1^*$ and the three-steady-state ordering), there exists $t' > 0$ such that $\bar{\pi}^{t'} < \bar{\pi}_2^*(\mu^0)$. Returning μ to μ^0 places the trajectory in the basin of attraction of $\bar{\pi}_1^*(\mu^0)$, yielding persistent convergence to $\bar{\pi}_1^*(\mu^0) < \bar{\pi}^0$. $\square$

The following Lemma is useful for proving Proposition 3.

Lemma 7. *For all $\bar{\pi} \in [0, 1]$ and $V \geq 0$, $\frac{dm_{\beta_0}(\bar{\pi})}{d\bar{\pi}} \leq \frac{d\theta(\bar{\pi})}{d\bar{\pi}}$, with strict inequality for all $\bar{\pi} > 0$ whenever $V > 0$ and $\beta_0 \in [0, 1)$.*

Proof. Differentiating m_{β_0} yields

$$\frac{dm_{\beta_0}}{d\bar{\pi}} = \frac{d\theta(\bar{\pi})}{d\bar{\pi}} \times \frac{(1 - \beta_0)[1 - (1 - \beta_0)\theta(\bar{\pi})V]}{[1 + (1 - \beta_0)\theta(\bar{\pi})V]^3}.$$

Set $z = (1 - \beta_0)\theta(\bar{\pi})V \geq 0$ and define

$$A(z) = \frac{(1 - \beta_0)(1 - z)}{(1 + z)^3},$$

so $\frac{dm_{\beta_0}}{d\bar{\pi}} = \frac{d\theta(\bar{\pi})}{d\bar{\pi}} \times A(z)$. I now show that $A(z) < 1$ for all $z > 0$. If $z \geq 1$, then $1 - z \leq 0$, so $A(z) \leq 0 < 1$. If $0 < z < 1$, then $(1 - \beta_0) < 1$ and $(1 - z) < 1 \leq (1 + z)^3$, so $A(z) < 1$. At $z = 0$, $A(0) = 1 - \beta_0 \leq 1$, with equality only when $\beta_0 = 0$. Hence, $A(z) < 1$ for all $z > 0$, i.e., whenever $\theta(\bar{\pi}) > 0$, $V > 0$, and $\beta_0 \in [0, 1)$. $\square$

Proof of Proposition 3.

Proof. Recall from the definition of the slope envelope that

$$\tilde{p}'(\bar{\pi}; V) = \frac{d\theta(\bar{\pi})}{d\bar{\pi}} \times \max_{\beta \in [0,1]} h(\beta, \bar{\pi}, V),$$

where $h(\beta, \bar{\pi}, V) = \frac{(1-\beta)|1-(1-\beta)\theta V|}{(1+(1-\beta)\theta V)^3}$. Setting $z = (1-\beta)\theta(\bar{\pi})V$, the maximum over $\beta \in [0, 1]$ is equivalent to

$$\max_{\beta \in [0,1]} h = \frac{1}{\theta(\bar{\pi})V} \max_{z \in [0, \theta(\bar{\pi})V]} g(z), \quad g(z) \equiv \frac{z|1-z|}{(1+z)^3}.$$

Let $w = \theta(\bar{\pi})V > 0$. If $w \geq 2 - \sqrt{3}$, the interior maximum of g at $z^* = 2 - \sqrt{3}$ is feasible and $g(z^*) = \frac{\sqrt{3}}{18}$, giving $\max_{\beta} h = \frac{\sqrt{3}}{18w}$. If $w < 2 - \sqrt{3}$, the maximum of g on $[0, w]$ is attained at $z = w$ (since g is increasing on $[0, 2 - \sqrt{3}]$), giving $\max_{\beta} h = \frac{g(w)}{w} = \frac{1-w}{(1+w)^3}$.

I show that $\max_{\beta} h$ is strictly decreasing in V for each $\bar{\pi} > 0$ in both regimes.

Case 1: $w \geq 2 - \sqrt{3}$. $\max_{\beta} h = \frac{\sqrt{3}}{18\theta(\bar{\pi})V}$, so $\frac{\partial}{\partial V} \max_{\beta} h = -\frac{\sqrt{3}}{18\theta(\bar{\pi})V^2} < 0$.

Case 2: $w < 2 - \sqrt{3}$. $\max_{\beta} h = \frac{1-\theta V}{(1+\theta V)^3}$, so

$$\frac{\partial}{\partial V} \max_{\beta} h = \frac{2\theta(\theta V - 2)}{(1 + \theta V)^4} < 0,$$

where the inequality holds because $\theta V < 2 - \sqrt{3} < 2$.

Continuity at the boundary $w = 2 - \sqrt{3}$ is verified by observing that both expressions equal $\frac{\sqrt{3}-1}{(3-\sqrt{3})^3}$ at $w = 2 - \sqrt{3}$.

As $\frac{d\theta}{d\bar{\pi}} > 0$ for $\bar{\pi} > 0$ and $\max_{\beta} h$ is strictly decreasing in V , the pointwise envelope $\tilde{p}'(\bar{\pi}; V) = \frac{d\theta}{d\bar{\pi}} \times \max_{\beta} h$ is strictly decreasing in V for each $\bar{\pi} > 0$. As the supremum of pointwise-decreasing functions is decreasing, $\tilde{P}'(V) = \sup_{\bar{\pi}} \tilde{p}'(\bar{\pi}; V)$ is weakly decreasing in V . The inequality is strict because, at any maximizer $\bar{\pi}^*(V)$ of $\tilde{p}'(\cdot; V)$, $\theta(\bar{\pi}^*) > 0$ (since the maximizer lies in the interior where $\theta' > 0$), so the pointwise strict decrease holds at the maximizer.

To verify that $\bar{\Delta}(V)$ is strictly decreasing, recall that $\bar{\Delta}(V) = \sup_{\bar{\pi}} \max_{\beta, \beta'} |m_{\beta}(\bar{\pi}; V) - m_{\beta'}(\bar{\pi}; V)|$. For fixed $\bar{\pi} > 0$ and any $\beta \neq \beta'$, the gap $|m_{\beta} - m_{\beta'}|$ is strictly decreasing in V because higher V compresses all m_{β} terms toward zero (as $m_{\beta}(\bar{\pi}; V) \rightarrow 0$ for each β as $V \rightarrow \infty$, with the rate depending on β , and the highest- m term decreases fastest). Taking the supremum preserves the strict decrease by the same maximizer argument.

Continuity of $\tilde{P}'(V)$ and $\bar{\Delta}(V)$ follows from continuity of the constituent functions and compactness of $[0, 1]$, proving statement (i).

For statement (ii), I show that three steady states persist for all $V \in (0, V')$ whenever they exist at $V = V'$. The argument proceeds in two steps: first establishing persistence of the three crossings in the one-dimensional aggregate map, then extending to the full two-dimensional system. Recall $g(\bar{\pi}; V) = \Psi(\bar{\pi}; V) - \bar{\pi}$, where $\Psi(\bar{\pi}; V) = 1 - \bar{\gamma}(1 - m_{\beta_0}(\bar{\pi}; V))$. I verify the four conditions that jointly ensure exactly three zeros of $g(\cdot; V)$ for all $V \in (0, V']$: (a) $g(0; V) = 1 - \bar{\gamma} > 0$, independent of V ; (b) $g(1; V) = -\bar{\gamma}(1 - m_{\beta_0}(1; V)) < 0$ for all $V > 0$, since $m_{\beta_0}(1; V) = \frac{(1-\beta_0)}{(1+(1-\beta_0)V)^2} < 1$; (c) the endpoint slopes

satisfy $\frac{d\Psi(0;V)}{d\bar{\pi}} = \bar{\gamma}(1 - \beta_0)\theta'(0) < 1$ (independent of V , using $\theta'(0) < 1$ by the sigmoidal assumption and $\bar{\gamma}(1 - \beta_0) < 1$) and $\frac{d\Psi(1;V)}{d\bar{\pi}} \leq \bar{\gamma}\theta'(1) < 1$ (where the first inequality follows from Lemma 7 and the second from the sigmoidal assumption); (d) by Lemma 7, $\frac{dm_{\beta_0}(\bar{\pi};V)}{d\bar{\pi}}$ is pointwise non-decreasing as V decreases, so $\sup_{\bar{\pi}} \frac{d\Psi(\bar{\pi};V)}{d\bar{\pi}} \geq \sup_{\bar{\pi}} \frac{d\Psi(\bar{\pi};V')}{d\bar{\pi}} > 1$ for all $V \leq V'$. Conditions (a)–(d) together imply that $g(\cdot; V)$ starts positive, decreases below zero (by (a), (c) at $\bar{\pi} = 0$, and (d)), rises above zero (by (d)), and ends negative (by (b), (c) at $\bar{\pi} = 1$). By the intermediate value theorem and the sigmoidal single-peaked structure of $\frac{d\Psi}{d\bar{\pi}}$, $g(\cdot; V)$ has exactly three zeros $\bar{\pi}_1^*(V) < \bar{\pi}_2^*(V) < \bar{\pi}_3^*(V)$ for every $V \in (0, V']$.

At $V = V'$, three steady states $\pi_k^*(V')$ of the full map $T(\cdot; V')$ exist by hypothesis. Each is a non-degenerate fixed point: $\det(DT(\pi_k^*; V') - I) \neq 0$, as established in the proof of Proposition 2(ii) via the implicit function theorem applied to the bias parameters $(\beta_0, \beta_1, \beta_2)$. By the implicit function theorem now applied to the parameter V , each $\pi_k^*(V)$ extends uniquely as a C^1 function of V in a neighborhood of V' . Loss of a steady state as V decreases from V' requires a bifurcation at which $\det(DT - I) = 0$, i.e., an eigenvalue of DT equals 1. In the unperturbed system ($\bar{\Delta} = 0$), the eigenvalues of DT at a crossing $\bar{\pi}_k^*$ are $\bar{\gamma} \frac{dm_{\beta_0}}{d\bar{\pi}} \big|_{\bar{\pi}_k^*}$ and 0, so the non-degeneracy gap at crossing k is $|1 - \bar{\gamma} \frac{dm_{\beta_0}(\bar{\pi}_k^*)}{d\bar{\pi}}|$. For $\bar{\Delta} > 0$, the eigenvalues are perturbed by $O(\bar{\Delta})$, so $\det(DT - I)$ remains nonzero as long as $\bar{\Delta}$ is small relative to the non-degeneracy gap. By the above argument, the three one-dimensional crossings are non-degenerate for all $V \in (0, V']$ (the slopes $\bar{\gamma} \frac{dm_{\beta_0}}{d\bar{\pi}}$ at the crossings are strictly different from 1 and vary continuously in V), so the minimum non-degeneracy gap

$$\sigma \equiv \inf_{V \in (0, V']} \min_{k \in \{1, 2, 3\}} \left| 1 - \bar{\gamma} \frac{dm_{\beta_0}(\bar{\pi}_k^*(V); V)}{d\bar{\pi}} \right|$$

is strictly positive by continuity and compactness of $[0, V']$ (the infimum extends by continuity to $V = 0$ where the three crossings also exist). As $\bar{\Delta}(V) \leq \bar{\Delta}(0) = \beta_0 < \infty$ is bounded, the non-degeneracy condition holds for all $V \in (0, V']$ whenever $\bar{\Delta}$ is small relative to σ , precisely the content of the maintained condition $\bar{\Delta} < \hat{\Delta}$ from Proposition 2(ii), whose threshold $\hat{\Delta}$ is bounded below by σ scaled by a Lipschitz constant (see footnote 28). Hence, no bifurcation occurs on $(0, V']$, the three two-dimensional steady states persist throughout, and $\Omega(V') \subset \Omega(V)$.

□

Proof of Proposition 4.

Proof. From the proof of Proposition A3,

$$\Psi(\bar{\pi}; V) = 1 - \bar{\gamma}(1 - m_{\beta_0}(\bar{\pi}; V)).$$

Steady states of the full system correspond to fixed points of $\Psi(\cdot; V)$ up to the perturbation $\mathcal{E}(\pi)$ bounded by $\bar{\Delta}$, as established in the proof of Proposition 2.

Differentiating $m_{\beta_0}(\bar{\pi}; V)$ with respect to V yields

$$\frac{\partial m_{\beta_0}}{\partial V} = - \frac{2(1 - \beta_0)^2 \theta(\bar{\pi})^2}{[1 + (1 - \beta_0)\theta(\bar{\pi})V]^3} < 0$$

for all $\bar{\pi} > 0$. Therefore

$$\frac{\partial \Psi(\bar{\pi}; V)}{\partial V} = \bar{\gamma} \frac{\partial m_{\beta_0}}{\partial V} < 0$$

for all $\bar{\pi} \in (0, 1]$. Define $g(\bar{\pi}; V) \equiv \Psi(\bar{\pi}; V) - \bar{\pi}$. At any stable steady state $\bar{\pi}_k^*(V)$, Ψ crosses the diagonal from above, so $\frac{\partial g(\bar{\pi})}{\partial \bar{\pi}} < 0$. As $\frac{\partial g(\bar{\pi})}{\partial V} < 0$ and $\frac{\partial g(\bar{\pi})}{\partial \bar{\pi}} < 0$ at $\bar{\pi}_1^*$, the implicit function theorem gives

$$\frac{d\bar{\pi}_1^*}{dV} = -\frac{\partial g/\partial V}{\partial g/\partial \bar{\pi}} < 0,$$

so $\bar{\pi}_1^*(V)$ is strictly decreasing in V .

From Lemma 7,

$$\frac{dm_{\beta_0}(\bar{\pi}; V)}{d\bar{\pi}} = \frac{d\theta(\bar{\pi})}{d\bar{\pi}} \times A(z), \quad z = (1 - \beta_0)\theta(\bar{\pi})V,$$

where $A(z) \rightarrow 0$ as $z \rightarrow \infty$. Fix $\delta > 0$. For all $\bar{\pi} \in [\delta, 1]$, $\theta(\bar{\pi}) \geq \theta(\delta) > 0$, so $z \geq (1 - \beta_0)\theta(\delta)V \rightarrow \infty$, and $A(z) \rightarrow 0$ uniformly on $[\delta, 1]$ as $V \rightarrow \infty$. On $[0, \delta]$, by the hypothesis $\tilde{p}'(0) < \bar{\gamma}^{-1}$ in Proposition 2(ii), we have $\bar{\gamma} \frac{dm_{\beta_0}}{d\bar{\pi}} \leq \bar{\gamma} \tilde{p}'(\bar{\pi}) < 1$ near $\bar{\pi} = 0$ for any $V \geq V^0$. As $\delta \rightarrow 0$ this region vanishes. Hence,

$$\lim_{V \rightarrow \infty} \sup_{\bar{\pi} \in [0, 1]} \frac{d\Psi(\bar{\pi}; V)}{d\bar{\pi}} = \lim_{V \rightarrow \infty} \bar{\gamma} \sup_{\bar{\pi} \in [0, 1]} \frac{dm_{\beta_0}(\bar{\pi}; V)}{d\bar{\pi}} = 0.$$

By Proposition 3, both $\tilde{P}'(V) = \sup_{\bar{\pi}} \tilde{p}'(\bar{\pi}; V)$ and $\bar{\Delta}(V) = \sup_{\bar{\pi}} [p(\bar{\pi}; V) - m_{\beta_0}(\bar{\pi}; V)]$ are strictly decreasing and continuous in V . By the above, $\tilde{P}'(V) \rightarrow 0$ as $V \rightarrow \infty$. As $p(\bar{\pi}; V) \rightarrow 0$ and $m_{\beta_0}(\bar{\pi}; V) \rightarrow 0$ as $V \rightarrow \infty$ for each $\bar{\pi}$, with $p(\bar{\pi}; V) \geq m_{\beta_0}(\bar{\pi}; V)$, the gap $\bar{\Delta}(V) \rightarrow 0$ as well. Therefore, there exists V' such that for all $V \geq V'$,

$$2\bar{\Delta}(V) + \tilde{P}'(V) < \gamma(\eta_2, \varepsilon)^{-1},$$

which is the uniqueness condition of Proposition 2(i). By that proposition, the full model admits a unique steady state $\bar{\pi}_1^*(V) \in [0, 1]^2$ for all $V \geq V'$. By Lemma 3, this steady state is not $(1, 1)$, so $\bar{\pi}_1^*(V) \in (0, 1)$. Moreover, $\bar{\pi}_1^*(V) \leq \bar{\pi}_1^*(V^0) < \bar{\pi}_2^*(V^0)$, proving statement (i).

Now fix $V'' \geq V'$ satisfying statement (i) and let $\bar{\pi}^0 \in (\bar{\pi}_2^*(V^0), 1)$. Under V'' , the unique steady state $\bar{\pi}_1^*(V'') < \bar{\pi}_2^*(V^0)$ is globally attracting by the contraction established in the proof of Proposition 2(i). Hence, the trajectory $\{\bar{\pi}^t\}_{t \geq 0}$ converges to $\bar{\pi}_1^*(V'')$ from $\bar{\pi}^0$. As $\bar{\pi}_1^*(V'') < \bar{\pi}_2^*(V^0)$ and convergence is guaranteed in finite time since the unique steady lies strictly below $\bar{\pi}_2^*(V_0)$, there exists a $t' < \infty$ such that $\bar{\pi}^{t'} < \bar{\pi}_2^*(V^0)$. Returning V to V^0 at time t' restores the original three-steady-state configuration. Because $\bar{\pi}^{t'} < \bar{\pi}_2^*(V^0)$, the trajectory now lies in the basin of attraction of $\bar{\pi}_1^*(V^0)$ under V^0 , and converges to $\bar{\pi}_1^*(V^0) < \bar{\pi}^0$, proving statement (ii). $\square$

Proof of Lemma 4.

Proof. Recall that $\Delta\pi_\tau = (1 - \eta_\tau)[1 - \bar{\rho}_\tau(\pi) - \pi_\tau] + \eta_\tau(1 - \pi_\tau)\varepsilon$. As $\bar{\rho}_\tau(\pi) \in [0, 1]$ and $\varepsilon \geq 0$, the expression is minimized at $\bar{\rho}_\tau = 1$ and $\varepsilon = 0$, giving $\Delta\pi_\tau \geq -(1 - \eta_\tau)\pi_\tau$. Therefore, $-\Delta\pi_\tau \leq (1 - \eta_\tau)\pi_\tau \leq 1 - \eta_\tau$, and

$$\bar{\pi}^t - \bar{\pi}^{t+1} = -(q\Delta\pi_1 + (1 - q)\Delta\pi_2) \leq q(1 - \eta_1)\pi_1 + (1 - q)(1 - \eta_2)\pi_2 \leq q(1 - \eta_1) + (1 - q)(1 - \eta_2).$$

Substituting $1 - \eta_1 = (1 - q)(1 + \mu)$ and $1 - \eta_2 = q(1 + \mu)$ yields $\bar{\pi}^t - \bar{\pi}^{t+1} \leq q(1 - q)(1 + \mu) + (1 - q)q(1 + \mu) = 2q(1 - q)(1 + \mu) = C^*(\mu, q)$. $\square$

The following Lemma corresponds to footnote 21.

Lemma 8. Under Ω with $\mu > -1$ and $\varepsilon > 0$, the unstable tipping point $\bar{\pi}_2^*(\mu, V)$ is strictly increasing in both μ and V .

Proof. From the proof of Proposition A3, $g(\bar{\pi}; \mu, V) \equiv \Psi(\bar{\pi}; \mu, V) - \bar{\pi}$ satisfies $\frac{\partial g(\bar{\pi}_2^*)}{\partial \bar{\pi}} > 0$ at the unstable tipping point (where Ψ crosses the diagonal from below). From the proof of Proposition A3, $\frac{\partial g}{\partial \mu} < 0$. From the proof of Proposition 4, $\frac{\partial g}{\partial V} < 0$. Applying the implicit function theorem to $g(\bar{\pi}_2^*; \mu, V) = 0$ in each argument separately yields

$$\frac{d\bar{\pi}_2^*}{d\mu} = -\frac{\partial g/\partial \mu}{\partial g/\partial \bar{\pi}} > 0, \quad \frac{d\bar{\pi}_2^*}{dV} = -\frac{\partial g/\partial V}{\partial g/\partial \bar{\pi}} > 0. \quad \square$$

Proof of Proposition 5.

Proof. By Proposition 4(i), under (V'', μ^0) the model admits a unique globally attracting steady state $\bar{\pi}_1^*(\mu^0, V'') < \bar{\pi}_2^*(\mu^0, V^0)$. Since the trajectory $\{\bar{\pi}^t\}$ converges to $\bar{\pi}_1^*(\mu^0, V'')$, it eventually falls strictly below both $\bar{\pi}_2^*(\mu^0, V^0)$ and $\bar{\pi}_2^*(\tilde{\mu}, V^0)$, so $T_V, \tilde{T}_V < \infty$.

By hypothesis, $\bar{\pi}^0 \geq \bar{\pi}_2^*(\tilde{\mu}, V^0) > \bar{\pi}_2^*(\mu^0, V^0)$, so $\tilde{T}_V \geq 1$. By definition of $\tilde{T}_V$ as a minimum, $\bar{\pi}^{\tilde{T}_V} < \bar{\pi}_2^*(\tilde{\mu}, V^0)$ and $\bar{\pi}^{\tilde{T}_V-1} \geq \bar{\pi}_2^*(\tilde{\mu}, V^0)$.

Suppose for contradiction that $\bar{\pi}^{\tilde{T}_V} \leq \bar{\pi}_2^*(\mu^0, V^0)$. Then

$$\bar{\pi}^{\tilde{T}_V-1} - \bar{\pi}^{\tilde{T}_V} \geq \bar{\pi}_2^*(\tilde{\mu}, V^0) - \bar{\pi}_2^*(\mu^0, V^0) > C^*(\mu^0, q),$$

where the second inequality is the assumption on $C^*(\mu^0, q)$. Lemma 4 applied under matching parameter μ^0 gives $\bar{\pi}^{\tilde{T}_V-1} - \bar{\pi}^{\tilde{T}_V} \leq C^*(\mu^0, q)$, a contradiction. Hence, $\bar{\pi}^{\tilde{T}_V} \in (\bar{\pi}_2^*(\mu^0, V^0), \bar{\pi}_2^*(\tilde{\mu}, V^0))$. In particular $\bar{\pi}^{\tilde{T}_V} > \bar{\pi}_2^*(\mu^0, V^0)$, so the trajectory has not yet crossed the lower threshold at time $\tilde{T}_V$, establishing $\tilde{T}_V < T_V$.

Under $(\tilde{\mu}, V^0)$, the maintained conditions ensure the three-steady-state structure holds. From the proof of Proposition A3, $\bar{\pi}_1^*(\mu, V)$ is strictly decreasing in μ , so $\bar{\pi}_1^*(\tilde{\mu}, V^0) < \bar{\pi}_1^*(\mu^0, V^0) < \bar{\pi}_2^*(\mu^0, V^0)$. As $\bar{\pi}^{\tilde{T}_V} < \bar{\pi}_2^*(\tilde{\mu}, V^0)$, the trajectory under $(\tilde{\mu}, V^0)$ initiated at $\bar{\pi}^{\tilde{T}_V}$ lies in the basin of attraction of $\bar{\pi}_1^*(\tilde{\mu}, V^0)$ and converges to it. As $\bar{\pi}_1^*(\tilde{\mu}, V^0) < \bar{\pi}_2^*(\mu^0, V^0)$, the trajectory eventually falls strictly below $\bar{\pi}_2^*(\mu^0, V^0)$, so $T_\mu < \infty$.

Under (V'', μ^0) , $T_V < \infty$ and by definition $\bar{\pi}^{T_V} < \bar{\pi}_2^*(\mu^0, V^0)$. Restoring V^0 at $t = T_V$ returns the model to the three-steady-state configuration under (μ^0, V^0) , with $\bar{\pi}^{T_V}$ lying strictly in the basin of attraction of $\bar{\pi}_1^*(\mu^0, V^0)$. The economy therefore converges persistently to $\bar{\pi}_1^*(\mu^0, V^0) < \bar{\pi}^0$.

As established above, $\bar{\pi}^{\tilde{T}_V} \in (\bar{\pi}_2^*(\mu^0, V^0), \bar{\pi}_2^*(\tilde{\mu}, V^0))$. At $t = \tilde{T}_V$, V is restored to V^0 and μ is raised to $\tilde{\mu}$. The maintained conditions ensure three steady states under $(\tilde{\mu}, V^0)$. As $\bar{\pi}^{\tilde{T}_V} < \bar{\pi}_2^*(\tilde{\mu}, V^0)$, the trajectory under $(\tilde{\mu}, V^0)$ lies in the basin of attraction of $\bar{\pi}_1^*(\tilde{\mu}, V^0)$ and converges to it. As $\bar{\pi}_1^*(\tilde{\mu}, V^0) < \bar{\pi}_2^*(\mu^0, V^0)$, the trajectory crosses $\bar{\pi}_2^*(\mu^0, V^0)$ in finite time T_μ , by the argument above. At $t = \tilde{T}_V + T_\mu$, both instruments are restored to (μ^0, V^0) . By the definition of T_μ , $\bar{\pi}^{\tilde{T}_V+T_\mu} < \bar{\pi}_2^*(\mu^0, V^0)$, so the trajectory lies strictly in the basin of attraction of $\bar{\pi}_1^*(\mu^0, V^0)$ and converges to it persistently. The strict inequality $\tilde{T}_V < T_V$ is established above, proving the proposition. $\square$

Proof of Proposition 6.

Proof. The maintained assumptions place the economy in the high-bias basin under (V^0, μ^0) , so the counterfactual trajectory $\{\bar{\pi}_{cf}^t\}$ converges to $\bar{\pi}_3^*(\mu^0, V^0)$. I consider each instrument in turn.

V-intervention. Under (V'', μ^0) , Proposition 4(i) implies a unique globally attracting steady state $\bar{\pi}_1^*(V'') < \bar{\pi}_2^*(V^0)$, so the intervention trajectory $\{\bar{\pi}_{int}^t\}$ converges to $\bar{\pi}_1^*(V'')$. Add and subtract $W(\bar{\pi}_{int}^t; V^0, \mu^0)$ in the total welfare comparison:

$$\begin{aligned} W(\bar{\pi}_{int}^t; V'', \mu^0) - W(\bar{\pi}_{cf}^t; V^0, \mu^0) &= \underbrace{W(\bar{\pi}_{int}^t; V'', \mu^0) - W(\bar{\pi}_{int}^t; V^0, \mu^0)}_{= B_V(t) \geq 0} \\ &\quad + \underbrace{\frac{1}{2} [\bar{S}(\bar{\pi}_{cf}^t; V^0, \mu^0) - \bar{S}(\bar{\pi}_{int}^t; V^0, \mu^0)]}_{\text{bias-reduction gain} \geq 0}. \end{aligned}$$

The first term satisfies $B_V(t) \geq 0$ for all t by the envelope argument in Section 5.2. For the second term, I claim $\bar{\pi}_{int}^t \leq \bar{\pi}_{cf}^t$ for all $t \geq 0$. At $t = 0$, $\bar{\pi}_{int}^0 = \bar{\pi}_{cf}^0 = \bar{\pi}^0$. For $t \geq 1$, the intervention trajectory converges monotonically to $\bar{\pi}_1^*(V'') < \bar{\pi}_2^*(V^0) < \bar{\pi}^0$, while the counterfactual converges to $\bar{\pi}_3^*(V^0) > \bar{\pi}_2^*(V^0)$, so $\bar{\pi}_{int}^t < \bar{\pi}_{cf}^t$ for all $t \geq 1$. Since $\bar{S}$ is increasing in $\bar{\pi}$ (as $f(z)$ is increasing in z and $z_k = (1 - \beta_k)\theta(\bar{\pi})V$ is increasing in $\bar{\pi}$ through θ), the bias-reduction gain is non-negative for all t and strictly positive for $t \geq 1$.

At $t = 0$, $B_V(0) > 0$ strictly: $V'' > V^0$, $\bar{\pi}^0 > 0$ (so $\theta(\bar{\pi}^0) > 0$), and $\frac{\partial u^*}{\partial V} > 0$ for every match type with $z > 0$. Hence, the total welfare difference is strictly positive for all $t \geq 0$.

μ -intervention. The decomposition established in Section 5.2 gives

$$W(\bar{\pi}_{int}^t; \tilde{\mu}) - W(\bar{\pi}_{cf}^t; \mu^0) = \underbrace{\frac{1}{2} [\bar{S}(\bar{\pi}_{cf}^t; \mu^0) - \bar{S}(\bar{\pi}_{int}^t; \mu^0)]}_{\text{bias-reduction gain}} - \underbrace{J_\mu(t)}_{\mu\text{-penalty}}.$$

At the start of the intervention $t = 0$, $\bar{\pi}_{int}^0 = \bar{\pi}_{cf}^0 = \bar{\pi}^0$, so the bias-reduction gain is zero. The μ -penalty satisfies $J_\mu(0) > 0$ (since $\bar{\pi}^0 > 0$ implies $\bar{s}_{out, \tau}^* - s_0^* > 0$ by the effort ordering $s_2^* > s_1^* > s_0^*$). Hence, the total welfare difference equals $-J_\mu(0) < 0$. Under $\tilde{\mu} > \mu^0$, the intervention trajectory falls faster than the counterfactual: $\bar{\pi}_{int}^t < \bar{\pi}_{cf}^t$ for all $t \geq 1$ (the higher contact rate generates more positive out-group interactions, reducing bias faster, while the counterfactual remains in the high-bias basin). The bias-reduction gain $\frac{1}{2} [\bar{S}(\bar{\pi}_{cf}^t; \mu^0) - \bar{S}(\bar{\pi}_{int}^t; \mu^0)]$ is therefore strictly positive and growing for $t \geq 1$, as $\bar{\pi}_{cf}^t$ converges upward to $\bar{\pi}_3^*$ while $\bar{\pi}_{int}^t$ converges downward toward $\bar{\pi}_1^*(\tilde{\mu})$. Simultaneously, $J_\mu(t)$ is declining (as shown in Section 5.2, bias reduction narrows the effort gap $\bar{s}_{out, \tau}^* - s_0^*$). Since the bias-reduction gain is eventually bounded below by $\frac{1}{2} [\bar{S}(\bar{\pi}_3^*; \mu^0) - \bar{S}(\bar{\pi}_1^*(\tilde{\mu}); \mu^0)] > 0$ and $J_\mu(t) \rightarrow J_\mu(\infty)$ evaluated at $\bar{\pi}_1^*(\tilde{\mu})$, which is bounded, there exists $\hat{t}$ at which the gain exceeds the penalty. For all $t > \hat{t}$, the total welfare difference is strictly positive.

At $t = T_M$, μ is returned to μ^0 . The μ -penalty drops to zero immediately: $J_\mu(t) = 0$ for $t \geq T_M$ because the matching parameter is μ^0 in both the intervention and counterfactual paths. The bias-reduction gain persists. By the maintained assumption, $\bar{\pi}_{int}^{T_M} < \bar{\pi}_2^*(\mu^0)$, so the post-intervention trajectory converges to $\bar{\pi}_1^*(\mu^0)$, while the counterfactual converges to $\bar{\pi}_3^*(\mu^0)$. Hence, $\bar{S}(\bar{\pi}_{cf}^t; \mu^0) > \bar{S}(\bar{\pi}_{int}^t; \mu^0)$ for all $t \geq T_M$, and the total welfare difference equals the bias-reduction gain, which is strictly positive and converges to ΔW as $t \rightarrow \infty$. $\square$

Proof of Proposition 7.

Proof. All three policies converge to the same low-bias steady state $\bar{\pi}_1^*(\mu^0, V^0)$, so the per-period welfare from period T_P onward (where T_P is the last period of intervention under policy P) converges to the

common value $W(\bar{\pi}_1^*(\mu^0, V^0))$. The present discounted welfare under any successful policy P can therefore be decomposed as

$$\mathcal{W}_P = \sum_{t=0}^{T_P-1} \delta^t [W^P(t) - c^P(t)] + \sum_{t=T_P}^{\infty} \delta^t W^P(t),$$

where $W^P(t)$ is per-period welfare and $c^P(t) \in \{c_V, c_\mu\}$ is the fiscal cost at time t along the trajectory induced by policy P .

Write the no-intervention counterfactual welfare as $\mathcal{W}_0 = \sum_{t=0}^{\infty} \delta^t W_{cf}(t)$, where $W_{cf}(t) \rightarrow W(\bar{\pi}_3^*)$ as $t \rightarrow \infty$. For each policy P , define the net welfare gain relative to the counterfactual:

$$\mathcal{W}_P - \mathcal{W}_0 = \underbrace{\sum_{t=0}^{\infty} \delta^t [W^P(t) - W_{cf}(t)]}_{\text{total transition-path welfare gain}} - \underbrace{\sum_{t=0}^{T_P-1} \delta^t c^P(t)}_{\text{total fiscal cost}}.$$

Because all policies converge to $\bar{\pi}_1^*(\mu^0, V^0)$ and the counterfactual converges to $\bar{\pi}_3^*(\mu^0, V^0)$, the tail of the transition-path welfare gain converges to $\Delta W = \frac{1}{2}[\bar{S}(\bar{\pi}_3^*) - \bar{S}(\bar{\pi}_1^*)]$ per period. The discounted value of this permanent gain, $R = \Delta W/(1 - \delta)$, is identical across all three policies.

The policies differ only in their transition-path welfare effects and fiscal costs during the intervention phase. Substituting the definitions of $B_V(t)$ and $J_\mu(t)$:

Policy V: The per-period welfare difference relative to the counterfactual is $B_V(t) + G_V(t)$, where $G_V(t) = \frac{1}{2}[\bar{S}(\pi_{cf}^t; \mu^0) - \bar{S}(\pi_V^t; \mu^0)]$ is the bias-reduction gain along the V -only trajectory, and the per-period fiscal cost is c_V . Hence, the welfare-adjusted cost in each period is $c_V - B_V(t) = \tilde{c}_V^V(t)$, and the total welfare-adjusted cost is C_V as defined in (23), up to the common bias-reduction gain that is absorbed into R .

Policy M: The per-period welfare difference relative to the counterfactual is $-J_\mu(t) + G_M(t)$, the per-period fiscal cost is c_μ , and the welfare-adjusted cost is $c_\mu + J_\mu(t) = \tilde{c}_\mu^M(t)$. The total is C_M as in (22).

Policy S: The welfare-adjusted costs are $\tilde{c}_V^S(t)$ during $[0, \tilde{T}_V)$ and $\tilde{c}_\mu^S(t)$ during $[\tilde{T}_V, \tilde{T}_V + T_\mu)$. The total is C_S as in (24).

Since R is common to all three policies, the net welfare gain satisfies $\mathcal{W}_P - \mathcal{W}_0 = R - C_P + \Xi_P$, where Ξ_P collects differences in the timing of the bias-reduction gain across policies that do not affect the long-run welfare. Because all trajectories converge to the same steady state, differences in Ξ_P across policies are of order $O(\delta^{T_P})$ and vanish relative to R for patient planners (δ close to 1). At any $\delta \in (0, 1)$, the ranking $\mathcal{W}_P > \mathcal{W}_{P'}$ holds if and only if $C_P < C_{P'}$, because the bias-reduction gains, while differing in their timing across policies, are second-order relative to the welfare-adjusted costs for planners who are sufficiently patient.

Strictly, for any $\delta \in (0, 1)$, define the *full* welfare-adjusted cost inclusive of transition-path differences:

$$C_P^* = \sum_{t=0}^{T_P-1} \delta^t c^P(t) - \sum_{t=0}^{\infty} \delta^t [W^P(t) - W_{cf}(t)].$$

Then $\mathcal{W}_P - \mathcal{W}_0 = -C_P^*$, and P is welfare-preferred to P' iff $C_P^* < C_{P'}^*$. During the intervention phase, $W^P(t) - W_{cf}(t)$ decomposes into the instrument-specific welfare effect (captured by B_V or $-J_\mu$) and the bias-reduction gain. The welfare-adjusted per-period costs $\tilde{c}_V(t)$ and $\tilde{c}_\mu(t)$ as defined in (21) incorporate

the instrument-specific welfare effects. The bias-reduction gains differ across policies only during the intervention phases (since all post-intervention trajectories converge to the same steady state), so $C_P^* = C_P - \Gamma_P$ where $\Gamma_P = \sum_t \delta^t G_P(t)$ is the discounted bias-reduction gain along policy P 's trajectory. Since Γ_P depends on the trajectory, pairwise rankings reduce to comparing $C_P - \Gamma_P$ across policies.

The pairwise comparisons in statements (i)–(iii) hold by construction: policy P is welfare-preferred to P' iff $C_P - \Gamma_P < C_{P'} - \Gamma_{P'}$. Because the proposition defines the ranking in terms of C_P (the welfare-adjusted costs as defined in (22)–(24)), which already incorporate the instrument-specific welfare effects through B_V and J_μ , the stated pairwise conditions $C_P < C_{P'}$ are necessary and sufficient whenever the bias-reduction gain differential $\Gamma_P - \Gamma_{P'}$ is negligible or can be absorbed into the cost comparison. This holds exactly when the comparison involves policies that share initial trajectory segments (as in statement (i), where V and S share the first $\tilde{T}_V$ periods), and holds to a close approximation otherwise, with the approximation improving as $\delta \rightarrow 1$. $\square$